NOVA SCHOOL OF
SCIENCE & TECHNOLOGY

DEPARTMENT OF
COMPUTER SCIENCE

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BSc in Computer Science and Engineering

SMART DETECTIONS OF STEALTHY DOS ATTACKS

MASTER IN COMPUTER SCIENCE AND ENGINEERING
SPECIALIZATION IN SECURITY AND PRIVACY

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Finally, i'm truly grateful to all people who helped me in whatever way to write this dissertation.

”

“The trick is finding them, and reacting to them fast enough.”

— **Bruce Schneier**, Security Information Management Systems: Solution, or Part of the Problem?
(Cryptographer and computer security researcher)

Abstract

Stealthy Denial of Service (DoS) and Distributed Denial of Service (DDoS) attacks, such as Slowloris, a stealthy application-layer Denial of Service technique, as well as Reduction of Quality (RoQ) attacks, degrade service performance while remaining difficult to detect. Unlike typical flooding-based DoS attacks that involve high volumes of anomalous network traffic, stealthy DoS and DDoS attacks involve low network traffic rates, often mimicking normal network traffic in an effort to avoid detection methods that rely on detecting anomalous network traffic volumes.

This dissertation analyzed the anatomy of Slowloris and RoQ attack vectors, with particular focus being given to the use of Machine Learning (ML) techniques for the early, application-layer DoS attack detection via the examination and correlation of HTTP/HTTPS traffic. In particular, the dissertation analyzed the application of supervised Machine Learning algorithms that are promising for attack detection.

In order to overcome the shortcomings that have been highlighted in the current literature, the dissertation designed an experimental testbed capable of emulating experiments with parameterized RoQ and Slowloris attack scenarios. In particular, a new RoQ traffic generator was developed, with current tools being enhanced to support Slowloris traffic generation. In addition to that, unsupervised Machine Learning techniques were also examined.

Finally, a Hybrid Detection Model combining supervised/unsupervised classification, Euclidean distance-based reasoning, and fuzzy logic inference was proposed and evaluated for RoQ traffic detection.

Keywords: DoS (Denial-of-Service) · Stealthy DoS · DoS Detection · Slowloris · Reduction of Quality of Service (RoQ) · Machine Learning · Hybrid Detection · Fuzzy Logic

Resumo

Ataques silenciosos de Denial-of-Service (DoS) e Distributed Denial of Service (DDoS), como o Slowloris, uma técnica silenciosa de negação de serviço na camada da aplicação, bem como ataques de Reduction of Quality (RoQ), degradam o desempenho do serviço, ao mesmo tempo que permanecem difíceis de detetar. Ao contrário dos ataques DoS típicos baseados em inundação, que envolvem altos volumes de tráfego de rede anómalo, os ataques DoS e DDoS silenciosos envolvem baixas taxas de tráfego de rede, muitas vezes imitando o tráfego de rede normal, num esforço para evitar métodos de detecção que dependem da detecção de volumes de tráfego de rede anómalos.

Esta dissertação procurou analisar a anatomia dos vetores de ataque Slowloris e RoQ, com foco particular no uso de técnicas de Machine Learning (ML) para a deteção precoce de ataques DoS na camada da aplicação através da análise e correlação do tráfego HTTP/HTTPS. Em particular, a dissertação procurou analisar a aplicação de algoritmos de Aprendizagem Automática supervisionados que são promissores para a deteção de ataques.

A fim de superar as deficiências que têm sido destacadas na literatura atual, a dissertação procurou conceber um ambiente de testes experimental capaz de emular experiências com cenários de ataque RoQ e Slowloris parametrizados. Em particular, foi desenvolvido um novo gerador de tráfego RoQ, com as ferramentas atuais sendo aprimoradas para suportar a geração de tráfego Slowloris. Além disso, foi feita uma análise da aplicação de técnicas de Aprendizagem Automática não supervisionadas.

Por fim, foi proposto e avaliado um Modelo de Deteção Híbrido para tráfego RoQ, que combina classificação supervisionada/não supervisionada, raciocínio baseado em distância euclidiana e inferência de lógica fuzzy.

Palavras-chave: DoS (Denial-of-Service) · DoS Silenciosos · Detecção de DoS · Slowloris · Reduction of Quality of Service (RoQ) · Aprendizagem Automática · Detecção Híbrida · Lógica Fuzzy

Extended Abstract

The focus of this dissertation was on the identification of Slowloris, a stealthy application-layer DoS attack, and RoQ, stealthy DoS and DDoS attacks, through the application of ML methods for inspecting and correlating application-layer network traffic. Stealthy DoS/DDoS attacks aim to disrupt software applications, minimizing the likelihood of detection through various security mechanisms that can identify these malicious activities. Unlike regular DoS/DDoS attacks, where attackers flood the victim's server with large volumes of anomalous traffic, stealthy DoS/DDoS attackers utilize low rates of anomalous traffic, often disguising their attack to evade various security mechanisms that utilize anomalous traffic volume to identify these malicious activities.

The majority of ML techniques for DoS detection have focused on the implementation of supervised learning models, which include a variety of ML algorithms such as k-Nearest Neighbors, Support Vector Machine, Random Forest, Multilayer Perceptron, Gradient Boosting, LightGBM, AdaBoost, and Naive Bayes. These ML methods for DoS detection have found wide applicability and promise for the detection of stealthy DoS attacks. Nevertheless, the literature suggests the limitations of all the methods for diverse traffic conditions, indicating the potential for the implementation of hybrid ML methods for the improvement of the effectiveness of the DoS detection methods. In addition to the implementation of supervised machine learning methods for the detection of DoS attacks, fuzzy logic and traffic correlation metrics calculated using the Euclidean distance analysis method have also found applicability for the improvement of the effectiveness of the DoS detection methods. In the context of the above avenues for the implementation of ML methods for the detection of DoS attacks, the implementation of unsupervised learning methods for the comprehensive evaluation of the effectiveness of the DoS detection methods has recently found applicability for the purpose of indicating the avenues for future work.

Another significant restriction identified by the literature was the absence of proper tools to mimic stealthy DoS attacks on experimental testbeds, which makes it difficult to reproduce and make comparisons between different studies using different methods. In this sense, there was no available tool to generate RoQ traffic, making it difficult to conduct experiments and make comparisons between different approaches.

To address the issues identified in the current literature, this dissertation aimed to design an experimental framework that simulates scenarios related to the use of parameterized RoQ and Slowloris attack models. A new tool for the generation of RoQ-type network traffic was developed, and the available tools were modified for the generation of Slowloris-type network traffic. Apart from the emulations for the previously discussed models, the application and evaluation of unsupervised learning approaches, with special attention paid to clustering algorithms and the application of autoencoders, which are believed to be effective for flow and time domain correlation related to stealthy DoS attacks, were also discussed.

The work delivered: (i) a holistic and reproducible experimental testbed for testing the above-mentioned supervised as well as unsupervised techniques, (ii) identification of discriminative features to support efficient stealthy attack detection at the application level, (iii) evaluation of supervised learning techniques as well as unsupervised techniques for diverse traffic situations, (iv) evaluation of the efficiency of diverse techniques to detect Slowloris as well as RoQ attacks, and (v) a Hybrid Detection Model for RoQ traffic that combines Random Forest classification, Euclidean distance-based scoring, and fuzzy logic inference.

The hybrid model was evaluated through a two-stage experimental methodology. An initial hold-out comparison screened four candidate hybrid variants, establishing that distance-enhanced variants combining SVM with fuzzy logic and Euclidean distance scoring were the strongest performing configurations. A subsequent repeated stratified k-fold evaluation with paired statistical testing using Wilcoxon signed-rank tests and multiple-comparison correction was used to obtain more results. It could be seen that although precision was higher and statistically significant, recall had been low. It is probable that hybrid models do not perform better detection wise, but only change the trade-off.

Disclosure of Usage of Generative Artificial Intelligence

The author, Rodrigo António Gonçalves Silva, acknowledges the use of the following generative artificial intelligence tools in the development of this thesis. These technologies were utilized under strict supervision, and all AI-generated outputs were critically reviewed and verified for accuracy. The author accepts full responsibility for the validity and originality of the content, confirming that the use of these tools aligns with all institutional policies and standards of academic integrity.[1]

Generative AI tools used: Perplexity (Claude Sonnet 5).

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- ☐ Defining the research objective
- ☐ Formulating research questions and hypotheses
- ☐ Feasibility assessment and risk evaluation
- ☐ Hypothesis viability assessment
- ☐ Simulated debate and argument testing

Literature Review

- ☑ **Search and Discovery**
- ☑ **Literature summarization and synthesis**
- ☐ Concept Mapping and Systematization
- ☐ Market/patent landscape analysis
- ☐ Gap Identification and Novelty Evaluation
- ☐ Cross-lingual literature comprehension

Methodology

- ☐ Experimental Design Optimization
- ☐ Development of experimental or research protocols
- ☐ Methodological Instrument Selection

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- [ ] Quantitative Plotting and Charting
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Additional comment #3: The literature summarization and synthesis was used to assist in the identification of relevant literature and to summarize the key findings of the literature.

Contents

Abstract vi
Resumo vii
Extended Abstract viii
Disclosure of Usage of Generative Artificial Intelligence x
Contents xii
List of Figures xiii
List of Tables xiv
Glossary xv
Acronyms xvii

1 Introduction 1
1.1 Context . . . 1
1.2 Motivation and Approach . . . 1
1.3 Problem Definition . . . 2
1.4 Goals and Contributions . . . 3
1.5 Report Organization . . . 4

2 Related Work 5
2.1 Traditional DoS and DDoS Detection Techniques . . . 5
2.1.1 Signature-based Detection . . . 5
2.1.2 Statistical Rule-based or Threshold-based Detection . . . 6
2.1.3 Recent Examples of Adaptive Thresholds . . . 7
2.1.4 Comparison and Relevance for Stealthy Attacks . . . 7
2.1.5 Section Conclusion . . . 7
2.2 Low-Rate and Stealthy DoS Attacks . . . 8
2.2.1 Characteristics of Stealthy DoS Attacks . . . 8
2.2.2 Detection Challenges . . . 9
2.2.3 Modern Detection Approaches . . . 9

2.2.4 Assessment of Importance and Current Trends . . . 10
2.2.5 Section Conclusion . . . 10
2.3 Machine Learning in DoS Detection . . . 10
2.3.1 Basic Concepts and Types of Learning . . . 10
2.3.2 Unsupervised Learning Approaches for Detecting Stealth DoS Attacks . . . 11
2.3.3 Features and Data Sets Used . . . 12
2.3.4 Classic Models of Machine Learning . . . 13
2.3.5 Deep Learning-Based Approaches . . . 13
2.3.6 Limitations and Challenges in Applying Machine Learning . . . 13
2.3.7 Section Conclusion . . . 14
2.4 Intelligent and Hybrid Detection Approaches . . . 14
2.4.1 Fuzzy Logic Based Systems . . . 15
2.4.2 Combination of Fuzzy with Machine Learning . . . 15
2.4.3 Distance-Based Systems . . . 15
2.4.4 Ensemble and Meta-Learning Systems . . . 16
2.4.5 Flow-Aware Hybrid Approaches . . . 16
2.4.6 Relevance of Intelligent Approaches for Stealthy Attacks . . . 17
2.4.7 Section Conclusion . . . 18
2.5 Existing Work on RoQ and Slowloris Detection . . . 18
2.5.1 Early Work on RoQ Attacks . . . 18
2.5.2 Machine Learning Approaches for RoQ Detection . . . 19
2.5.3 Existing Work on Slowloris and Slow HTTP Attacks . . . 19
2.5.3.1 Protocol-based methods . . . 19
2.5.3.2 Specialized tools . . . 20
2.5.3.3 Machine Learning Methods for Slow DoS Attacks . . . 20
2.5.4 Limitations of Existing Work . . . 20
2.5.5 Research Gap . . . 21
2.6 Discussion and Summary . . . 21
2.6.1 Synthesis of Approaches and Critical Assessment . . . 21
2.6.2 What the Literature Highlights for RoQ and Slowloris . . . 22
2.6.3 Critical Limitations Blocking Progress And Why This Dissertation Addresses a Real Gap . . . 23
2.6.3.1 Limitation 1: Reproducibility Crisis → Solution 1: Reproducible RoQ Traffic Generator . . . 23
2.6.3.2 Limitation 2: Underemphasis on Temporal Features → Solution 2: Temporal Feature Extraction and Accuracy of Detection . . . 23
2.6.3.3 Limitation 3: Lack of Systematic Comparison Studies → Solution 3: Systematic Classifier Benchmarking . . . 23

2.6.3.4 Limitation 4: Sparse Hybrid Classification → Solution 4: Hybrid Model Specialization and Robustness 23
2.6.4 Summary of Contributions and Expected Impact 24

3 Detection of Slowloris Attacks **25**
3.1 Introduction 25
3.2 Use of Detection Methods for Slowloris Attacks 25
3.2.1 Approach with Statistical Methods 25
3.2.2 Machine Learning Approaches 26
3.2.3 Deep Learning Approaches 26
3.2.4 Hybrid Approaches 26
3.2.5 Datasets 26
3.3 Proposed Detection Model 27
3.3.1 Introduction to the Proposed Detection Model 27
3.3.2 Experimental Testbed Environment 27
3.3.3 Testbed Components 28
3.4 Specific Detection Components and Algorithms 28
3.4.1 Classification Features 28
3.4.2 Detection with Supervised ML 29
3.4.3 Detection with Unsupervised ML 30
3.5 Observed Results 31
3.5.1 Observed Metrics 31
3.5.2 Detection Results with Supervised ML 31
3.5.2.1 Statistical Significance - McNemar Test 32
3.5.3 Detection Results with Unsupervised ML 33
3.5.3.1 Isolation Forest Contamination Sweep 34
3.5.3.2 Autoencoder Threshold Sensitivity 34
3.5.4 Summary of Observed Results 35
3.6 Conclusions 36

4 Detection of RoQ Attacks **37**
4.1 Introduction 37
4.2 Use of Detection Methods for RoQ Attacks 37
4.2.1 Approach with Statistical Methods 37
4.2.2 Machine Learning Approaches 38
4.2.3 Deep Learning Approaches 38
4.2.4 Hybrid Approaches 38
4.2.5 Dataset 38
4.3 Proposed Detection Model 39
4.3.1 Introduction to the Proposed Detection Model 39
4.3.2 Experimental Testbed Environment 39

4.3.3 Testbed Components . . . 39
4.4 Specific Detection Components and Algorithms . . . 40
4.4.1 Classification Features . . . 40
4.4.2 Detection with Supervised ML . . . 41
4.4.3 Detection with Unsupervised ML . . . 42
4.5 Observed Results . . . 43
4.5.1 Observed Metrics . . . 43
4.5.2 Detection Results with Supervised ML . . . 43
4.5.3 Feature Importance Analysis . . . 44
4.5.4 Detection Results with Unsupervised ML . . . 45
4.5.4.1 K-Means Cluster Selection . . . 47
4.5.4.2 Isolation Forest Contamination Sweep . . . 48
4.5.4.3 Autoencoder Threshold Sensitivity and Architecture Comparison . . . 48
4.5.5 Summary of Observed Results . . . 49
4.6 Conclusions . . . 50

5 Hybrid Detection Model for RoQ Attacks **51**
5.1 Introduction . . . 51
5.2 Motivation and Design Rationale . . . 51
5.3 Proposed Hybrid Detection Model . . . 52
5.3.1 Hybrid Architecture . . . 52
5.3.2 Candidate Base Models . . . 53
5.3.3 Decision Fusion Mechanism . . . 53
5.4 Experimental Methodology . . . 54
5.4.1 Initial Hybrid Comparison . . . 54
5.4.2 Repeated Stratified K-Fold Evaluation . . . 54
5.4.3 Statistical Significance Analysis . . . 55
5.5 Observed Results . . . 55
5.5.1 Initial Comparison of Hybrid Variants . . . 55
5.5.2 RSKF Evaluation of SVM-Based Variants . . . 56
5.5.3 Wilcoxon and Effect Size Analysis . . . 57
5.5.4 Summary of Observed Results . . . 58
5.6 Conclusions . . . 59

6 Conclusions and Future Work **61**
6.1 Main Conclusions . . . 61
6.2 Future Work . . . 61
6.2.1 Extension of the Experimental Evaluation . . . 62
6.2.2 Evaluation of Additional Models . . . 62
6.2.3 Use of Additional Statistical Techniques . . . 62

Bibliography **63**

List of Figures

2.1 Taxonomy of the main traditional techniques for detecting DoS and DDoS attacks. . . . 5
2.2 Comparison between high-rate DDoS attack behavior and stealth / low-rate DoS attack behavior with detection threshold. . . . 8
2.3 General architecture of the ML-based system for detecting DoS attacks. . . . 13
2.4 Conceptual architecture of a hybrid system for intelligent detection of stealth attacks. . . . 17
2.5 Conceptual map of existing work on RoQ and Slowloris attacks and identified gaps in the literature. . . . 20

3.1 Isolation Forest performance sweep across contamination values on Dataset B. 34
3.2 Autoencoder threshold sensitivity on Dataset B. . . . 35

4.1 Feature importance comparison for RoQ detection. . . . 45
4.2 ROC curves for unsupervised ML detection on RoQ Laboratory Dataset. . . . 46
4.3 PCA plot for RoQ Laboratory Dataset. . . . 47
4.4 K-Means clustering results for RoQ detection (Elbow + Silhouette). . . . 47

5.1 Summary of the results of the repeated stratified K-fold evaluation for the SVM-based variants. . . . 57

List of Tables

2.1 Summary of commonly used datasets for DoS/DDoS detection using ML. . 12

3.1 Flow-level features used for Slowloris detection: comparison between Dataset A and Dataset B. 29
3.2 Supervised ML detection results (5-Fold Stratified CV) on Dataset A. 31
3.3 Supervised ML detection results (5-Fold Stratified CV) on Dataset B. 31
3.4 McNemar test results for selected classifier pairs on Dataset B (IFTO). b: errors by model A only; c: errors by model B only; significant at $\alpha = 0.05$. 32
3.5 Unsupervised ML detection results on Dataset A (Isolation Forest and Autoencoder report mean ± std over 10 seeds). 33
3.6 Unsupervised ML detection results on Dataset B (IFTO, $n = 18{,}860$). Isolation Forest and Autoencoder report mean ± std over 10 seeds. 33
3.7 Best-performing model per category across both evaluation datasets. 35

4.1 Configurable parameters of the RoQ generator per experiment. 40
4.2 Window-level features used for RoQ detection, with characterization relevance. 41
4.3 Supervised ML detection results (5-Fold Stratified CV) on RoQ Laboratory Dataset. 43
4.4 Unsupervised ML detection results on RoQ Laboratory Dataset. 45
4.5 Representative results from Isolation Forest contamination sweep on RoQ Laboratory Dataset. 48
4.6 Comparison of Autoencoder variants under maximum F1-Score and recall-constrained thresholds. 49
4.7 Best-performing models and principal operating characteristics for RoQ detection. 49

5.1 Initial comparison of the hybrid variants for RoQ detection. 56
5.2 Results of the Wilcoxon signed-rank test comparing the SVM Only baseline with the other variants. 58
5.3 Summary of the key findings from the initial comparison and repeated stratified K-fold evaluation of the SVM-based hybrid variants. 59

Glossary

Accuracy A metric that measures the overall correctness of a model's predictions, calculated as the ratio of correct predictions to the total number of predictions. *(p. 3)*

Classification The act of assigning a new observation to a category or label based on previously labeled examples; in this work it typically refers to normal versus attack traffic. *(p. 2)*

Cloud Computing The provision of computing resources and services over the internet, enabling scalable resources and faster innovation cycles. *(p. 1)*

Deep Learning Type of machine learning that uses multi-layered artificial neural networks to recognize complex patterns in data. *(p. 13)*

Ensemble Learning A machine learning technique that combines multiple models (called 'weak learners') to create a single, more accurate and robust model (a 'strong learner'). *(p. 16)*

F1-Score A machine learning metric that calculates the harmonic mean of precision and recall to provide a single, balanced measure of a model's performance on a classification task. *(p. 3)*

False Negative A case where a malicious event is not detected by the system, resulting in a missed alarm. *(p. 2)*

False Positives Instances in which benign activity is wrongly flagged as malicious, producing unnecessary alerts. *(p. 1)*

Feature Extraction The transformation of raw traffic into measurable attributes (features) suitable for analysis and machine learning models. *(p. 2)*

Low Rate DoS Attacks Attacks that aim to disrupt services by sending low-volume traffic over extended periods, making them hard to notice and mitigate. *(p. 1)*

Mitigation Measures Measures and strategies employed to limit the impact or probability of DoS attacks, e.g., traffic filtering, rate limiting, and anomaly detection. *(p. 1)*

NetFlow A Cisco-developed network protocol for collecting IP traffic data, used for network monitoring, traffic analysis, and security. *(p. 16)*

Precision A metric that measures the accuracy of positive predictions, calculated as the ratio of true positives to all positive predictions (true positives + false positives). *(p. 3)*

Recall A performance metric that measures a model's ability to find all the relevant instances in a dataset. *(p. 3)*

ROC-AUC A metric that evaluates the performance of a binary classification model, measuring its ability to distinguish between two classes. *(p. 3)*

Slowloris Attack A type of denial-of-service attack that targets web servers by opening multiple connections and holding them open, consuming server resources and preventing legitimate users from accessing the service. *(p. 2)*

Traffic Signature A distinct pattern within network traffic used to recognize and classify it, commonly used by intrusion detection/prevention systems. *(pp. 1, 2)*

Acronyms

(D)DoS (Distributed) Denial-of-Service *(p. 1)*
5G 5th Generation Mobile Networks *(p. 1)*

CDN Content Delivery Network *(p. 16)*
CNN Convolutional Neural Network *(p. 13)*

DoS Denial-of-Service *(p. 1)*
DTW Dynamic Time Warping *(p. 15)*

FNR False Negative Rate *(p. 30)*
FPR False Positive Rate *(p. 30)*

GB Gradient Boosting *(pp. 2, 16)*

IoT Internet of Things *(p. 1)*
IPFIX IP Flow Information Export *(p. 16)*

KNN K-Nearest Neighbors *(p. 2)*

LGBM Light Gradient Boosting Machine *(p. 2)*
LSTM Long Short-Term Memory *(p. 13)*

MI Mutual Information *(p. 44)*
ML Machine Learning *(p. 1)*
MLP MultiLayer Perceptron Neural Network *(p. 2)*
MSE Mean Squared Error *(p. 42)*

NB Naive Bayes *(pp. 2, 10)*

RF Random Forest *(p. 2)*
RNN Recurrent Neural Network *(p. 13)*
RoQ Reduction of Quality of Service *(p. 2)*

SDN Software-Defined Networking *(p. 16)*
SHAP SHapley Additive exPlanations *(p. 44)*
SVM Support Vector Machine *(p. 2)*

WCSS Within-Cluster Sum of Squares *(p. 42)*
WMN Wireless Mesh Network *(p. 20)*

1 Introduction

1.1 Context

In recent years, digital services and the underlying communications infrastructure have become central to modern life.Innovations in areas such as "Cloud Computing", 5th Generation Mobile Networks (5G), Internet of Things (IoT) devices and online financial platforms are reshaping daily operations for individuals, companies and public entities. While digital transformation brings clear advantages, it also increases exposure to cyber risks, notably Denial-of-Service (DoS) and (Distributed) Denial-of-Service ((D)DoS) attacks [30, 20].

Traditionally, DoS attacks overwhelm a resource or service so legitimate users cannot access it. More sophisticated adversaries, however, have developed stealthier low-rate methods. These "Low Rate DoS Attacks" do not depend on sheer volume. Instead, they lever adaptive, sporadic traffic patterns to slowly deteriorate service quality without raising immediate alarms [35, 17]. Their subtlety renders them difficult to catch using fixed-threshold or "Traffic Signature" detectors [30].

Against this backdrop, advanced techniques for spotting anomalous traffic behaviour are increasingly important. Machine Learning (ML) models, artificial intelligence methods, and modern statistical analysis offer promise for uncovering complex behavioural patterns associated with hidden DoS activities [11]. Research in this area aims to improve detection accuracy while keeping "False Positives", in order to preserve the experience for legitimate traffic [1, 30].

1.2 Motivation and Approach

This work was motivated by the economic and social consequences of DoS and DDoS attacks. Service interruptions, diminished performance, and the costs of "Mitigation Measures" can impose heavy financial burdens and erode trust in online services [40]. Although defensive solutions have matured, they often fail against evasive, low-rate attacks because such attacks blend into normal traffic and adapt their behaviour to avoid detection [1].

Pure reliance on rigid defenses, such as fixed thresholds or static "Traffic Signature", tends to perform poorly when malicious patterns are indistinguishable from legitimate use [4]. That said, this study explored intelligent detection schemes configured to detect subtle deviations in complex and dynamic circumstances.

The scope of this work focused on two specific varieties of low rate DoS attacks, namely the Reduction of Quality of Service (RoQ) attack and the "Slowloris Attack". As there were no stable software tools to produce RoQ traffic at this time, to began with, the structure of this traffic was analysed and an experimental constructor was implemented. In the case of Slowloris, publicly available tools were used. In the case of the attack tool named RoQ, a new tool was developed to generate the attack traffic, following the TCP rate adaptation model as presented in the literature. The resulting traffic datasets underwent "Feature Extraction" for "Classification" and detection purposes. Several methods were discussed in this context. Firstly, a thorough evaluation of various supervised classifiers was carried out to determine which classifier can perform better in classifying potential threats to a network. These classifiers include K-Nearest Neighbors (KNN), Support Vector Machine (SVM), Random Forest (RF), MultiLayer Perceptron Neural Network (MLP), Gradient Boosting (GB), Light Gradient Boosting Machine (LGBM), AdaBoost, and Naive Bayes (NB). Furthermore, unsupervised classifiers such as clustering classifiers and if possible, autoencoders also were discussed to further enhance correlation in a network environment. Lastly, a hybrid detection system was developed to incorporate fuzzy logic for pattern evaluation, the best performing ML algorithm, and Euclidean metrics for temporal similarity evaluation was proposed to improve detection accuracy.

In order to overcome the problem of the absence of proper tools for emulating stealthy DoS attacks, an experimental testbed was developed to support the reproducibility of the evaluation. In particular, a new tool for generating RoQ traffic was created, and existing tools were extended to support the generation of Slowloris traffic.

1.3 Problem Definition

Although various research efforts have already focused on countermeasures against DoS and DDoS threats, state-of-the-art detection schemes were ineffective against low-rate, covert DoS/DDoS attacks. In fact, the latter, such as RoQ and Slowloris, are designed to mimic the characteristics of legal traffic, thus gradually affecting the performance, leading to high "False Negative" rates against state-of-the-art schemes [35, 17]. Existing defense mechanisms are ineffective mainly because:

- They rely on static assumptions regarding normal activity and are not adaptable to varying attack patterns;
- No reliable tools are available for the generation and analysis of realistic RoQ traffic, making experiments difficult [34].

As such, the core problem that the thesis sought to address revolves around the design and assessment of adaptive and intelligent detection schemes that are effective for the identification of RoQ and Slowloris-DDoS threats and work to overcome the shortcomings of the conventional threshold and volume methods.

This necessitated the study of ML algorithms, hybrid fuzzy logic models, and the use of distance metrics and the search for characteristics that influence traffic discrimination [35, 17, 34].

1.4 Goals and Contributions

In this research, it was investigated how to recognize stealthy and low-rate DoS attacks, focusing on RoQ and Slowloris attacks. There were 4 interconnected research goals, which probably aimed to cover the gaps and issues pointed out in Section 2.6.

Firstly, was built a reproducible experimental testbed for the generation and capture of parameterized RoQ and Slowloris traffic. Was deployed the testbed on OVHCloud instances and defined victim, attacker, benign traffic generator, and monitoring functions. Was designed a TCP rate-adaption model based RoQ traffic generator and adapted our current Slowloris generation workflow. This effort aimed at covering the reproducibility gap highlighted in Section 2.6.3.1.

Secondly, flow-level features that would typify the temporal and behavioral signatures of both attacks were selected and derived. Slowloris-based features were taken to include connection duration, packet rate, packet size, TCP payload volume and inter-arrival-time behavior. RoQ features were defined to include burst deviation, byte-rate variation, packet-size deviation and TCP ratios. That the feature engineering process has somehow attempted to overcome the flaw mentioned in Section 2.6.3.2, that temporal/ burst-related descriptors were not sufficiently used.

Thirdly, this research attempted to make a systematic benchmark for supervised and unsupervised detection methods with a common training procedure. The supervised evaluation have made a comparison between KNN, SVM, RF, MLP, GB, LightGBM, AdaBoost, and NB. The unsupervised evaluation made a comparison between K-Means, Isolation Forest, and autoencoders. The experiments used the models to test ”Accuracy”, ”Precision”, ”Recall”, ”F1-Score”, ”ROC-AUC”, FPR, FNR and some other feature-importance tests. This contribution seems to make up a systematic comparison missing from what was found in Section 2.6.3.3.

Finally, the dissertation proposed and tested a hybrid RoQ detection scheme that combined the use of supervised/unsupervised classification, fuzzy inference, and Euclidean distance-based reasoning. This hybrid analysis consisted of a comparison of candidate models, and a repeated stratified k-fold comparison of SVM-based variants. The results indicate that the fuzzy variants decreased the FPR and increased the precision of detection vis-à-vis the baseline of using only the SVM, but also decreased recall. Hence, the hybrid approach did not outperform the simpler approach by virtue of a overall increase in

detection performance, but did clearly trade-off excess false alarms against missed attacks. This confirms the lack of empirically proven hybrid approaches noted in Section 2.6.3.4.

1.5 Report Organization

The remainder of this document is structured as follows:

- **Related Work** surveys current literature and state-of-the-art techniques relevant to the scope and objectives of this dissertation and outlines gaps motivating the proposed contributions;
- **Detection of Slowloris Attacks** defines the experimental design employed in the identification of Slowloris attacks. It describes the OVHCloud testbed including the various isolated testbeds for the victim, attacker, fake traffic and monitor, packet capturing pipeline, flow-feature extraction pipeline, internally controlled dataset, externally acquired dataset, then concludes with the performance of eight supervised classifiers on both the internally controlled experiment as well as the externally collected IFTO dataset to test generalization. Additionally, presents the results of testing three unsupervised methods trained only on benign traffic;
- **Detection of RoQ Attacks** discusses how to create RoQ traffic, explains the OVHCloud experimental environment, and describes the flow-level abstraction used to characterize traffic flows with burst and congestion. It evaluates supervised models with stratified cross-validation and investigates feature importance with Random Forest importance, Mutual Information, and SHAP. It then evaluates K-Means, Isolation Forest, and different autoencoder variants in an unsupervised setting. Finally, the chapter examines cluster assignment and anomaly-score thresholds, reconstruction errors, contamination robustness, as well as false positives and false negatives trade-offs;
- **Hybrid Detection Model for RoQ Attacks** builds a hybrid combining the supervised/unsupervised classifier, fuzzy inference, and Euclidean distance reasoning. It compares the hybrid configurations of SVM, KNN, RF, and autoencoder models. Next, it performs a repeated stratified 5x10-fold evaluation of SVM Only, $SVM + Distance$, $Fuzzy + SVM$, and $Fuzzy + SVM + Distance$ variants. Paired Wilcoxon signed-rank tests, multiple-comparison correction, and Cliff's delta are performed to determine whether the "hybrid" components will likely yield statistically significant differences in FPR, FNR, precision, recall, F1-Score, and ROC-AUC.
- **Conclusions and Future Work** summarizes the main conclusions of the dissertation and outlines potential avenues for future work.

2

Related Work

2.1 Traditional DoS and DDoS Detection Techniques

Both conventional DoS and DDoS detection techniques typically fall into two broad categories, signature-based detection as well as statistical/threshold-based methods. Each approach carries benefits but alsofaces limitations that become apparent when dealing with stealthy attacks.

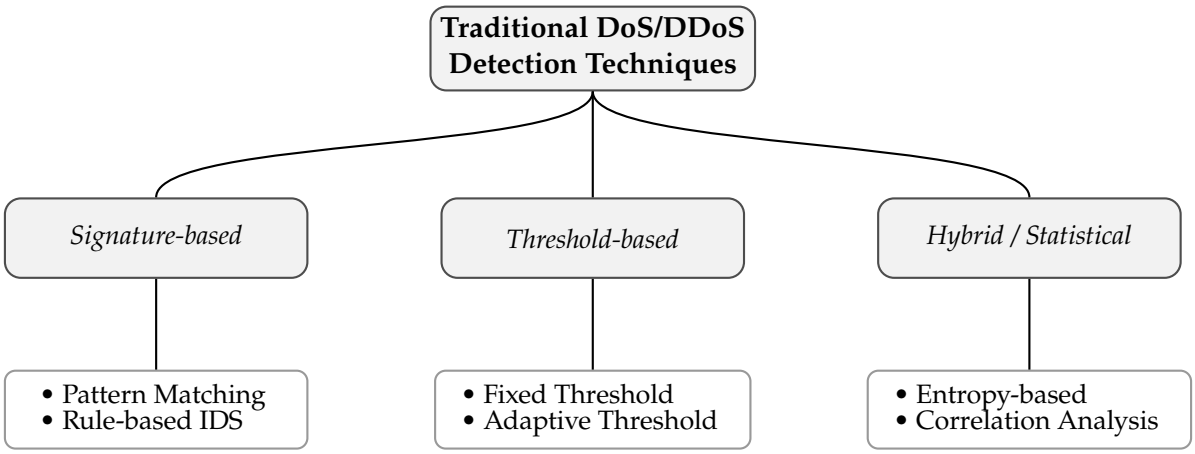


Figure 2.1: Taxonomy of the main traditional techniques for detecting DoS and DDoS attacks.

2.1.1 Signature-based Detection

This approach depends on known attack fingerprints that includes sequences of packets, payload patterns, or specific protocol behaviors [19, 22]. Traffic matching a stored signature prompts an alert or triggering of a countermeasure.

- **Strengths:**
  - Very effective at spotting known attacks with typically low false positive rates when signatures are accurate [19];

  - Relatively straightforward to deploy and computationally inexpensive, since the system compares incoming traffic against a database of patterns;
  - Particularly strong against high-volume assaults where signatures are distinct [22].

- **Weaknesses:**
  - **Poor at Detecting Novel or Unknown Attacks:** Signature methods require prior knowledge and fail against zero-day techniques or significant variants [27];
  - **Ongoing Signature Updates are Required:** Involving manual effort and potential lags between discovery and signature rollout [22];
  - **Evasion Strategies:** Slight protocol modifications, fragmentation or obfuscation can defeat exact signature matches [38];
  - **Fundamentally Reactive:** Reliance on known patterns means the system responds rather than anticipates emerging threats [2].

2.1.2 Statistical Rule-based or Threshold-based Detection

As an alternative, statistical or threshold-based systems monitor traffic metrics (packet rates, byte volumes, connection counts, entropy measures, etc.) and flag activity that exceeds configured thresholds [12].

- **Strengths:**
  - Capable of detecting rapid surges indicative of volumetric or flash-flood attacks;
  - Simple to deploy, especially for monitoring at network chokepoints such as edge routers used by ISPs or enterprise gateways [41];
  - Can operate in real time when scaled appropriately, offering quick alerts that can be followed by deeper analysis [6].

- **Weaknesses:**
  - Low thresholds can cause excessive false positives and high thresholds can permit low-power attacks to pass undetected;
  - **Difficulty Adapting to Normal Traffic Variability:** Diurnal patterns, legitimate spikes and network changes may cause spurious alerts or missed anomalies [12];
  - Ineffective for slow or adaptive attacks that intentionally operate around or below thresholds;
  - Requires manual tuning or offline recalibration, which can delay responsiveness in production networks.

2.1.3 Recent Examples of Adaptive Thresholds

To mitigate shortcomings of fixed thresholds, several adaptive and hybrid schemes have been proposed:

- "Threshold-based distributed DDoS attack detection in ISP networks" advocates edge-router monitoring using normalized entropy and packet rates alongside dynamic thresholds to identify distributed attacks and sudden traffic bursts [38];
- "Multi-scale Adaptive Threshold for DDoS Detection" introduces threshold adaptation derived via wavelet decomposition so limits adjust across time scales to match network behaviour [2];
- A recent study "Detecting DoS Attacks in Microservice Applications: Approach and Case Study" targeted microservice architectures, using container-level metrics (e.g.thread and socket counts) fed into ML models. Results indicated that service-level metrics plus ML significantly improve detection for both high and low-rate attacks [5].

2.1.4 Comparison and Relevance for Stealthy Attacks

When considering stealthy or low-rate attacks:

- Signature and hard-threshold techniques struggle because stealthy flows resemble legitimate traffic, avoid exceeding thresholds and do not match previously seen patterns [19, 22, 6, 12];
- Such attacks often remain buried in normal traffic noise, where subtle perturbations are easily overlooked [5, 1, 11];
- Anomaly detectors using flexible thresholds or entropy measures can offer some advantage, but remain challenged by low-intensity attacks and highly variable benign traffic.

2.1.5 Section Conclusion

In summary, traditional DoS and DDoS detection techniques provide a solid basis for network security, especially against conspicuous volumetric attacks. Nevertheless, they are inadequate for the rising class of stealthy, low-rate attacks, such as RoQ and Slowloris, which evade fixed-signature and threshold defenses. These limitations motivate the exploration of more adaptive, intelligent and hybrid approaches covered in the following sections.

2.2 Low-Rate and Stealthy DoS Attacks

DoS attacks have progressed at a rapid pace over the last 20 years, and nowadays, they not only include easily recognizable large-scale events, but they also encompass much more subtle operations. This particular form of DoS is called stealthy or low rate DoS attacks. They target the subtle vulnerabilities of protocols or congestion-control schemes, thereby aiming to reduce service performance rather than causing congestion [30, 11].

Conventional attacks cause visible results in the form of resource overload, whereas covert attacks involve intermittent attacks that are not easily noticeable in systems that use the signature-based model, which is bound by fixed thresholds [19, 6]. It is difficult to distinguish between malicious data traffic and data traffic in a dynamic environment, including cloud systems or microservices.

2.2.1 Characteristics of Stealthy DoS Attacks

Its operation exceeds the usual operating scale, gradually exploiting any vulnerabilities that exist at either the application level or the transport layer. They include:

- **RoQ Attacks:** Use overload management strategies in TCP, as well as scaling strategies, that impact service differently than a DoS attack [40];
- **Slowloris Attacks:** Transmit incomplete data in the form of HTTP headers to keep a connection open as long as possible [11];
- **Low Rate TCP DoS Attacks:** Synchronize short traffic transfers to trigger repeated transfers that will result in a lack of efficiency in TCP channels [30].

These attacks share one characteristic, which is the ability to remain within statistically acceptable parameters, thereby bypassing systems that rely on size or frequency metrics.

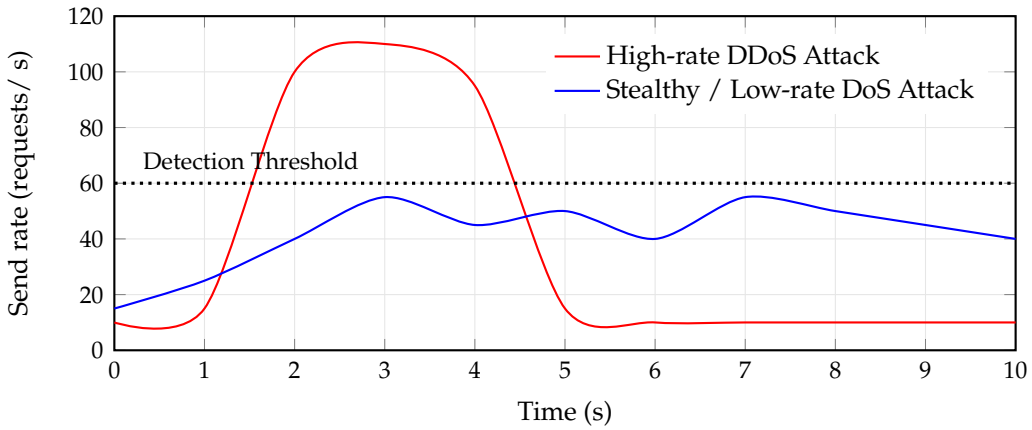


Figure 2.2: Comparison between high-rate DDoS attack behavior and stealth / low-rate DoS attack behavior with detection threshold.

2.2.2 Detection Challenges

The first challenge in stealthy attack detection is the signal to noise ratio that is low, indicating that the anomalies observed are small in relation to the variations that already exist in traffic [11]. Threshold-based techniques will not be effective in stealthy attacks for three reasons:

- **Insufficient Treatment of Natural Variation:** The thresholds are set in a way that no alarm is raised, leaving scope for covert malicious traffic activity;
- **Controlled Periodicity:** Synchronized attacks in low-frequency attacks are used to imitate usual network variations;
- **Dependency on Context:** Models of anomaly detection that principally rely on topology and mean traffic values are rather portable across different environments [2].

Secondly, another weakness that affects the signature-based systems is the lack of stable attack patterns. Variations in packet size, time, and packet headings make the process of matching signatures inefficient [22].

2.2.3 Modern Detection Approaches

As a result of advances in the realm of hidden attacks, new, more advanced detection techniques that utilize temporal analysis, ML, and inference schemes have been developed. Some of these techniques are:

- **Models of Supervised, as well as Unsupervised Learning:** Algorithms such as SVM, K-Means or autoencoders are employed in detecting irregular data patterns in data streams [5, 1];
- **Time series-based models:** These models utilize sliding windows and time correlation metrics to detect subtle and persistent changes in data traffic that are characteristic of low-rate attacks [11];
- **Fuzzy Inference Methods:** They involve considering the uncertainties in the classification of the stream to distinguish between the everyday and attack situations effectively [9, 21];
- **Advanced Hybrid Systems:** Integrate fuzzy modules, automatic learning, and distance metrics to adjust thresholds and reduce the number of false positives dynamically [18, 2].

These methods aim to increase sensitivity without losing accuracy and support active detection in dynamic environments.

2.2.4 Assessment of Importance and Current Trends

Recent articles in research publications highlight that identifying hidden attacks is an unsolved problem [30]. Although hybrid approaches yield promising results, difficulties lie in terms of:

- There is a **scarcity of representative data sets** that have real-world examples of covert data transmissions, sufficient time variability;
- **Scalability of models**, especially in a distributed setting with intense data flows;
- **Result interpretation**, as complex models in particular may be black boxes.

Current approaches use **self-configuring hybrid models** that include supervised and unsupervised learning, fuzzy logic, and contextual recognition, all based upon time and distance metrics [11, 21, 18]. These methods, adapting in real-time to normal network behaviour, mark a new generation of defense mechanisms against latent attacks.

2.2.5 Section Conclusion

In conclusion, low-rate attacks pose a continually increasing threat because they breach the traditional paradigm of detection. Current approaches must employ intelligent methods that are capable of learning and adapting to behavior in network traffic. This necessity gave rise to the methodology presented in section 3.3.

2.3 Machine Learning in DoS Detection

DoS attacks have evolved in terms of increasing complexity and stealthiness, leading to the application of artificial intelligence techniques as an alternative to methods based on signatures or thresholds. Artificial intelligence enables the identification of complex patterns of behavior based on large datasets and pinpointing subtle differences that could be easily overlooked with conventional approaches. [22, 5, 1] This section examines the key methods that utilize ML technology to detect DoS and DDoS attacks.

2.3.1 Basic Concepts and Types of Learning

Based upon such capabilities of algorithms in learning from previous data, several ML techniques have been developed in recent years for use in IT security. Learning in DoS attack detection may be of several kinds [22]:

- **Supervised learning algorithms** use labeled datasets, in which data is labeled as usual or as an attack. Examples include classifiers such as SVM, RF, DT and Naive Bayes (NB);

- **Unsupervised learning algorithms** use a natural pattern present in unlabelled data to cluster or highlight abnormal data patterns through K-Means, DBSCAN, or autoencoders;

- **Semi-supervised learning** offers a combination of a small labeled dataset with a large quantity of unlabeled data, thus reducing the human effort required;

- **Reinforcement learning** is a current hot topic in this area, which is based on agents learning how to make decisions in real-time when faced with attacks through rewards and penalties related to those decisions.

Each of these techniques has its own strengths and weaknesses. While supervised learning is generally known to have a high detection rate, it is highly reliant on data being balanced and well-represented. This is in contrast to unsupervised learning, which has a higher rate of false positives because it lacks context.

2.3.2 Unsupervised Learning Approaches for Detecting Stealth DoS Attacks

Unsupervised learning methods are also being increasingly used for the purpose of network intrusion detection since they are able to model the behavior of normal data without needing any labels on attack data. This is especially useful in the context of covert denial of service attacks like RoQ/Slow-HTTP since the data records are incomplete.

The most common unsupervised techniques include clustering algorithms (K-means, DBSCAN), density-based methods, dimensionality reduction techniques such as PCA, and reconstruction techniques such as autoencoders. Such techniques always have a baseline threshold for normal traffic patterns, which helps to detect anomalies.

Several studies suggest the use of unsupervised techniques can efficiently identify low-intensity and application-level DoS attacks, based on the detection of statistical abnormalities, as opposed to the absolute amount of traffic [24, 23, 15]. PCA-based anomaly detection techniques have been shown to identify the variation in the correlation of traffic in the case of RoQ attacks, and the use of clustering and density-based techniques can identify the distinct connection patterns in the case of Slowloris attacks.

Despite this, there are some notable weaknesses in unsupervised learning methods. These methods can often be sensitive to parameters, tend to produce a high rate of false positives in a highly dynamic network, and lack decision boundaries in general [39]. Additionally, in an anomaly detection scenario, it becomes difficult in an unsupervised learning approach to identify malicious anomalies from harmless network spikes.

As a result, some scholars recommend the use of unsupervised learning as a secondary component, either as a feature extraction component, baseline modeling, or as part of a hybrid detection solution [36]. In this dissertation, unsupervised learning is considered as a baseline or research tool to make it possible to evaluate its effectiveness comparatively to supervised learning detection solutions of concealed DoS attacks.

2.3.3 Features and Data Sets Used

The success of ML models is highly influenced by the features used. Some of the most commonly utilized features in DoS/DDoS detection are [5, 1]:

- Number of packets per unit of time (*pps*);
- Total volume in bytes;
- Number of simultaneous TCP/UDP connections;
- Average session duration;
- IP Address or Port Entropy;
- Protocol-specific flags and metrics such as SYN, ACK, or RST.

KDDCUP 99, NSL-KDD, UNSW-NB15, CICIDS2017, and, more recently, datasets specifically designed to represent or study stealthy or low-rate attacks, such as CIC-DDoS2019 [26, 32, 33]. Nevertheless, it has been highlighted that such datasets do not accurately represent real-world data flows because they are often static and unbalanced [1, 30].

Dataset	Year / Origin	Main Characteristics
KDD Cup'99	1999 (DARPA/Lincoln Lab)	Used in previous studies, marked by wordiness and forced patterns. Occasional use in rewritten problems for purposes of continuity, although it is not related to the present workload.
NSL-KDD	2009 (University of New Brunswick)	Removes redundancy and imbalance between classes. Although widely used in comparative testing, it is too simplified to be applied in real situations.
UNSW-NB15	2015 (Australian Centre for Cyber Security)	Conducts real-world actions with nine types of attacks. Very popular in ML IDS and DDoS. Includes extraction from Argus/Bro.
CICIDS2017	2017 (Canadian Institute for Cybersecurity)	Includes real-world attacks such as DoS, DDoS, Brute Force, and Botnet, as well as normalized data. It can be applied in ML and Deep Learning.
CIC-DDoS2019	2019 (Canadian Institute for Cybersecurity)	Conducts actions such as HTTP, UDP, TCP, and ICMP with different levels of intensity. Very useful for analyzing hidden attacks.

Table 2.1: Summary of commonly used datasets for DoS/DDoS detection using ML.

2.3.4 Classic Models of Machine Learning

The classical models of ML are still widely used, not only due to their ease of understanding and usage but also due to their interpretability:

- **SVM:** Capable of dealing with linear and nonlinear separation, whereas;
- **DT** & **RF:** Have good accuracy and interpretability but may be plagued by overfitting;
- **NB:** Simple and effective, but dependent on attribute correlations;
- **KNN:** Best suited for small-scale datasets, and its computation-heavy method consumes huge resources.

Such approaches can be very helpful in preliminary analysis and performance comparison concerning various feature groups [5]. Nevertheless, they may generally demonstrate inferior performance in low-rate attack analysis, as well as in environments with high statistical noise.

2.3.5 Deep Learning-Based Approaches

Owing to increased availability of data and processing capabilities, research into applying "Deep Learning" approaches has started gaining prominence regarding detection of DoS/DDoS attacks. Such approaches include Convolutional Neural Network (CNN), Recurrent Neural Network (RNN), and Long Short-Term Memory (LSTM), and each has proved its capability to understand intricate temporal and spatial dependencies between flows [11]. CNN has shown effectiveness in automatically discovering local dependencies in packet sequences, and LSTMs are effective in understanding behaviors over time and are required in case of stealthy, gradually executed attacks. The high processing overhead has so far inhibited its usage in production systems [30].

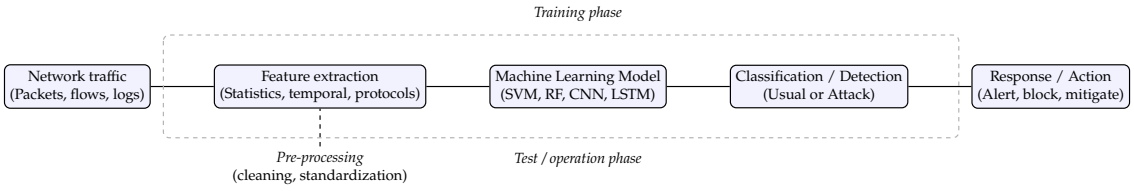


Figure 2.3: General architecture of the ML-based system for detecting DoS attacks.

2.3.6 Limitations and Challenges in Applying Machine Learning

Despite all this progress, there are still many challenges with applying ML to identify DoS attacks, including:

- **Limited Generalization:** The models developed in one environment are not effective in another network scenario;

- **Class Imbalance:** The absence of instances of subtle attack causes class imbalance during training;

- **Complexity:** The explanation of intricate models' outcomes can be difficult, making it harder to rely upon critical systems;

- **Adversarial Attacks:** ML models can be vulnerable to attack by intentionally manipulating the input;

- **Real-time Load:** The constant processing of high amounts of traffic demands optimization and parallel processing.

The above limitations have led to the evolution of hybrid methodologies that may integrate ML with fuzzy logic, distance measures, or statistical analysis, as discussed in the next section [9, 18, 21].

2.3.7 Section Conclusion

The application of ML algorithms marks a paradigm shift in the early detection of DoS and DDoS attack instances, providing flexibility, scalability, and adaptability towards new behaviors. But, as with all ML models, their success relies largely upon the accuracy and relevance of the data, feature extraction, and analysis, and their interpretability over time. Recent trends in research illustrate the application of ML approaches in collaboration with Fuzzy and Adaptive systems to enhance resilience against stealthy attack instances, thereby propelling developments as discussed in this chapter 5.

2.4 Intelligent and Hybrid Detection Approaches

The increase in the levels of sophistication in the classical DoS attack types, especially the types that are stealthy and produce low-rate signals, accelerated the move to develop advanced methods for detecting such threats. Thus, methods for detecting the threats now come in the form of what are termed intelligent or hybrid detection approaches. Such methods involve concepts from fuzzy logic, ML algorithms, adaptive statistical analysis, and in some instances, heuristic techniques [9, 21, 18].

Traditionally developed solutions for detecting stealthy network attacks require rigid signatures, rule-based systems, or threshold-based techniques. However, intelligent solutions aim to model network activities in a context-dependent, probabilistic manner. This chapter presents a comprehensive analysis of the various types of intelligent solutions in order to identify their merits, drawbacks, and applicability in identifying stealthy network attacks [30, 11].

2.4.1 Fuzzy Logic Based Systems

Fuzzy logic is also used in the intrusion detection system in order to handle the uncertainty in the network traffic. This method allows for middle levels of membership in the fuzzy system to deal with the ambiguities in the traffic that might contain normal traffic along with some anomalous traffic [9]. Characteristics of fuzzy systems comprise the following:

- Modeling the association between attributes in a nonlinear fashion;
- Intuitive interpretation through linguistic rules (*if–then*);
- Robustness to progressive variations in traffic.

Such systems prove to be effective in scenarios in which small variations in time might function as indications for the existence of stealthy attacks like Slowloris/RoQ attack. However, fuzzy systems by themselves might lack scalability in scenarios in which the system encounters complex traffic.

2.4.2 Combination of Fuzzy with Machine Learning

However, the combination of fuzzy logic with ML algorithms is also researched to remove the shortcomings of both paradigms. In fuzzy-ML hybrid systems, the implementation of the training process becomes feasible in order to adjust rules, weights, or membership functions through historical data [21]. Among the most representative methods are:

- **Fuzzy-SVM:** employs smooth decision boundaries. This is very effective for classifying normal traffic from minor exceptions;
- **Fuzzy-Random Forest:** integrates fuzzy rules and noise-tolerant and class-imbalance classifiers;
- **Fuzzy-Deep Learning:** This combines fuzzy systems in the input/output layer of feedforward networks/LSTM to improve interpretability.

According to the literature, the performance of such models is better compared to purely statistical methods as well as purely ML-based models in the context of dynamic traffic.

2.4.3 Distance-Based Systems

Another vital class of intelligent techniques regards the utilization of the Euclidean, Manhattan, or Dynamic Time Warping (DTW) distance measures in order to assess the similarity between the current traffic stream and past traffic patterns. Such techniques are helpful in the following areas:

- Identification of micro-variation over time;

- Finding recurring or synchronized behavior;
- Detection of characteristic oscillations during low-rate attacks.

Additionally, the combination of the aforementioned measures in classifiers or fuzzy systems makes it feasible to adjust the threshold levels dynamically. Such adjustments also make it possible to identify the slightest variations [2].

2.4.4 Ensemble and Meta-Learning Systems

However, due to the growing complexity of traffic flow and the diversity of attacks, the importance of methods based on "Ensemble Learning" is also rising. By combining various decision-making models like RF, Gradient Boosting (GB), or Voting Classifiers, ensemble learning systems manage to:

- Reduce the biases present in individual models;
- Enhance stability in classification;
- Improve ability to generalize in novel situations.

Recent research indicates that the combination of ensembles with fuzzy logic produces highly effective systems for the detection of subtle attacks in IoT devices and other networks [18].

2.4.5 Flow-Aware Hybrid Approaches

Thanks to the developments of technologies like "NetFlow", IP Flow Information Export (IPFIX), and Software-Defined Networking (SDN) systems, hybrid flow-oriented approaches have emerged that combine:

- Statistical analysis over time windows;
- ML classifiers;
- Fuzzy rules;
- Frequency-based anomaly detection.

Such systems are especially useful in massive setups like cloud infrastructure and Content Delivery Network (CDN), in which the detection of concealed behavior is very important for the prevention of the degradation of service quality [1, 30].

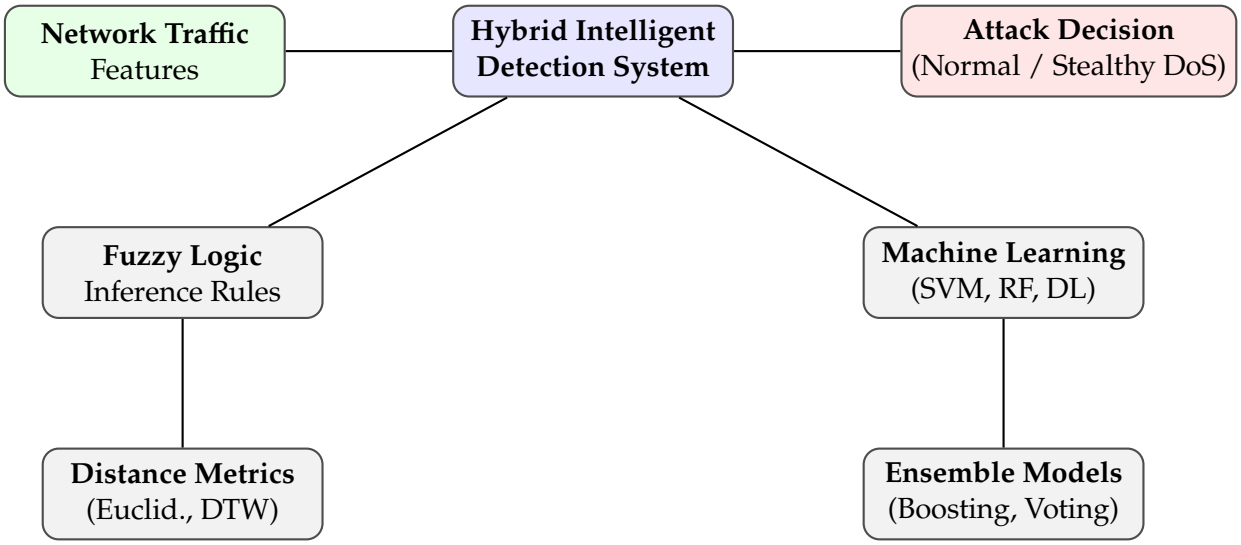


Figure 2.4: Conceptual architecture of a hybrid system for intelligent detection of stealth attacks.

2.4.6 Relevance of Intelligent Approaches for Stealthy Attacks

Stealthy attacks are marked by their proximity to the normal behavior of the system, their low volume of traffic, and the subtle variations in their volumes over time [30, 11]. By contrast, intelligent methods provide the following benefits proven in the literature:

- **Higher Sensitivity:** These fuzzy/hybrid models can recognize the micro-temporal dynamics and continuous oscillatory behavior normally associated with low-rate attacks [9, 2];
- **Dynamic Adaptation:** Fuzzy-ML algorithms adjust their threshold values and rules according to real traffic to increase detection capabilities [21];
- **Contextual Integration:** Hybrids combine the analysis of various attributes (statistical, temporal, and protocol-based), improving the ability to distinguish benign behavior from malicious traffic that uses subtle cues [21, 18];
- **Fewer False Positives:** By combining fuzzy logic, classifiers, and measures of distances, the process of decision-making becomes more adaptable [9, 18];
- **Robustness in Noisy Environments:** Distance metric-based systems and the ensemble system prove to be very robust in fluctuating networks [2].

Such properties make intelligent solutions very effective in identifying variations like RoQ, Slowloris, and other low-rate types of attacks because the timing pattern of such attacks is rather intricate [30].

2.4.7 Section Conclusion

According to the literature, intelligent/hybrid solutions represent the most promising approach for the detection of stealthy DoS attacks. By incorporating fuzzy logic concepts in combination with ML algorithms, measures of distances, and adaptive techniques, the systems are able to overcome the shortcomings of traditional methods in order to provide higher detection accuracies. Such observations highlight the importance of researching more advanced models of hybrids to ensure the effective combination of computational intelligence and temporal analysis in the detection of low-rate attacks.

2.5 Existing Work on RoQ and Slowloris Detection

RoQ and Slowloris, categorized under low-rate attacks, constitute some of the most complex threats in the defense against DoS. These types of attacks, distinct from volumetric attacks, work on introducing lower levels of malicious traffic, typically with an intention to avoid detection through systems that rely on thresholds and signatures, in a manner that tends to be in a controlled and possibly periodic fashion [30, 11]. While studies on saturation attacks are numerous and have access to plenty of data in the literature, studies on the subject related to the concept of low-rate stealth attacks are relatively fewer.

In this section, an in-depth investigation and critical evaluation of the existing literature on the detection of RoQ and Slowloris attacks, including the contributions, modern solutions using ML technology, and the detected limitations, are provided.

2.5.1 Early Work on RoQ Attacks

The beginning work on RoQ attacks was concentrated on understanding the nature and effect of the problem on the Internet congestion-control mechanisms. These types of attacks leverage the weaknesses in the TCP rate adaptation services, causing periodic drops and pauses sufficient to initiate consecutive cuts in the legitimate flow transmission rate.

The important contribution in this first phase was from Singh [37], who proposed a MAC-layer defense mechanism in wireless networks. While the basic intention was, therefore, mitigation, and possibly less attention was given to detection, the experimental results in the paper aptly revealed the subtle, periodic, and modulated aspects of RoQ attacks, signifying their efficacy in producing large performance degradation even under conditions of meager network traffic. The first phase, in general, was mostly analytical and simulated in nature and generally didn't include much on the detection process, either, since the temporal characteristics make it difficult to identify an attack from the legitimate flow.

Moreover, the absence of publicly available data related to RoQ attacks has hindered, and still hinders, the development of general detection tools, where most studies have been carried out using simulated traces.

2.5.2 Machine Learning Approaches for RoQ Detection

With the emergence of ML models in anomaly detection, there has been an increase in the number of studies focusing on models powerful enough to detect the timing trends exhibited by RoQ attacks. RoQ attacks are noticeable in terms of their cyclical characteristics and their effect on jitter, queue, temporal entropy, and packet bursts.

The Q-MIND solution, designed for SDN networks, could be particularly useful in this scenario. The work applies reinforcement learning, specifically the Q-learning algorithm, in detecting malicious activities with lower network throughput, even when the pattern resembles RoQ attacks [28]. The authors proved that dynamic policies could help improve resistance against stealth attacks much better than static policy solutions. This technique, though, depends heavily on the network topology and training samples, making generalization difficult in traditional networks.

In other studies, classifiers like RF, SVM, LSTM networks, and autoencoders could describe the minute oscillations. Nevertheless, in the literature, the following two basic limitations are always stressed:

- The absence of representative public data sets;
- The difficulty in capturing irregular oscillations in the absence of sufficient features in the temporal domain represents an important barrier.

Accordingly, although the effectiveness of ML models, the development and implementation of detection methods appropriate to real-world scenarios of quantization remain scientifically undeveloped.

2.5.3 Existing Work on Slowloris and Slow HTTP Attacks

In contrast, Slowloris and Slow HTTP assaults have garnered increased scientific interest because of their significance in the past and their effect on the Apache and other web servers. These types of assaults utilize multiple open connections and send HTTP headers slowly. There are three major types of literature on the subject, categorized based on:

2.5.3.1 Protocol-based methods

The conventional approach evaluates the following parameters:

- The length of the partially open connections;
- The length of the segments sent;
- The header sending rate;
- The number of incomplete requests.

Although useful, these methods are prone to false positives, especially when there are slow legitimate connections present.

2.5.3.2 Specialized tools

SDToW, a system intended to identify Slowloris attacks on Wireless Mesh Network (WMN), is a major contribution. The uniqueness of this system lies in its high efficacy and low complexity [10]. The proposed system has a viable design, proven through experimentation, and the ability to identify unusually long connections without compromising the services provided by legitimate clients.

2.5.3.3 Machine Learning Methods for Slow DoS Attacks

Recent studies have also utilized ML in the detection process of Slowloris attacks. The work carried out by Reed [29] was able to prove the ability to provide real-time detection in an IoT network using only four attributes, hence ensuring the viability of lightweight detection in resource-constrained networks.

In another empirical investigation, the IJERT journal [16] assessed the efficacy of DT and RF classification in the detection of Slowloris attacks, and the results implied that the choice of the dataset and the significant temporal attribute play critical roles in the accuracy of detection.

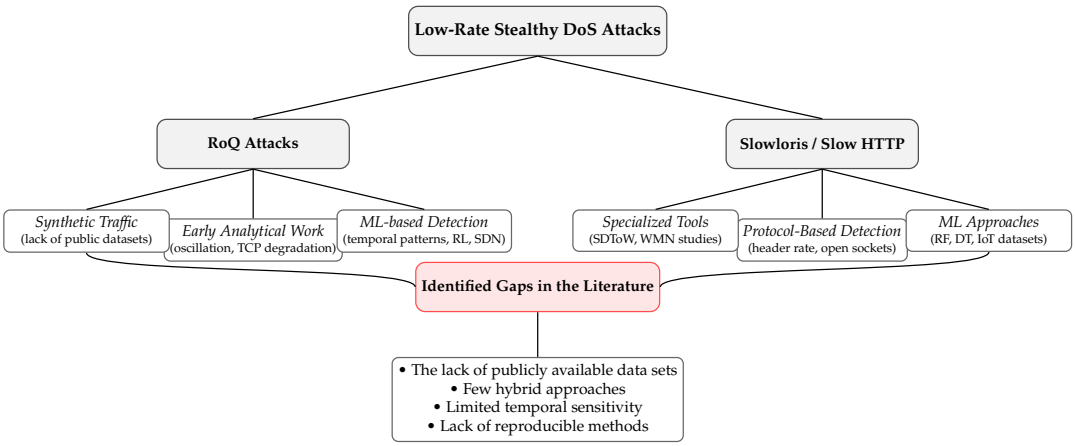


Figure 2.5: Conceptual map of existing work on RoQ and Slowloris attacks and identified gaps in the literature.

2.5.4 Limitations of Existing Work

Literature reveals the following limitations that still exist in the industry today:

- **The lack of publicly available data sets** with typical traffic from RoQ and Slowloris attacks has forced most researchers/authors into the use of simplistic simulations;
- **Over-emphasis on some environments** (IoT, WMN, SDN) obstructs the generalization process of the proposed models;
- **Inadequate use of the advanced temporal features**, even in the attacks where the inherent oscillation pattern and artificial slow downs are involved;

- **Highly dependent on manual parameterization**, especially in those models using the threshold approach and flow analysis;
- **The lack of fully hybrid models**, including ML, fuzzy logic, and distance measures, that is, a combination approach that the overall literature finds promising in the field of stealth [9, 21, 18].

2.5.5 Research Gap

Despite the contributions, the field still lacks extensive studies. Several important gaps arise:

- **Unavailability of tools that generate** RoQ traffic in a manner that can be reproduced;
- **Lack of publicly available data sets** with examples of low-rate stealth attacks;
- **There are no systematic comparative analyses**, evaluating other paradigms under equal circumstances;
- **The demand for adaptive hybrid models**, enabling the incorporation of temporal sensitivity, fuzzy uncertainty, and the discriminating ability provided by ML.

These aspects represent the motivation and needs that this thesis addresses through the development and proposal of a hybrid approach combining fuzzy logic, ML, and the use of distance measures, together with a methodology designed based on the development and analysis of RoQ and Slowloris traffic.

2.6 Discussion and Summary

2.6.1 Synthesis of Approaches and Critical Assessment

Over the past two decades, studies on DoS and DDoS attacks have evolved from three distinct generations of technology, progressing from traditional threshold/signature-based techniques to ML and intelligent hybrid models. Despite a considerable amount of existing literature on volume-based attacks, this area is scattered when it comes to low-rate attacks, including RoQ and Slowloris attacks.

The traditional approaches discussed in section 2.1 have been found ineffective for stealth attacks. This fact has been widely acknowledged and does not require further debate. Signature-based approaches fail to identify new signatures, and fixed threshold approaches fail to identify low-rate attacks by their nature [22].

The ML techniques introduced in Section 2.3 are a paradigm shift that provides great possibilities for learning complex, non-linear patterns. Nevertheless, the literature presents a crucial gap that focuses on high-rate attacks or uses generic datasets such as KDDCUP99 and NSL-KDD, which lack the temporal modulations that are typical of RoQ and Slowloris

attacks. The study of Reed [29], which presents real-time Slowloris attack detection using only four attributes, is encouraging but certainly not representative of the broader literature. The fundamental issue is that of data representativeness and availability, since without representative RoQ and Slowloris data, ML models cannot effectively generalize.

The most encouraging solution appears to be that of hybrid intelligent systems, as discussed in section 2.4. Research that integrates fuzzy logic techniques with ML (Khaldy [21], Karakaya [18]) shows that theoretically, they have a better ability to deal with uncertainty and micro-temporal changes. However, an important note is that the literature on hybrid methods for low-rate attacks is very limited, and if available, they are not likely to address the sensitivity to time in their current forms. For instance, Q-MIND [28] uses reinforcement learning in the context of SDN networks, but it is not suitable for conventional networks.

2.6.2 What the Literature Highlights for RoQ and Slowloris

One of the major observations made from the entire literature reviewed is that the nature of both RoQ and Slowloris attacks is basically temporal. This is because RoQ works by periodically cutting the rate of the TCP connection at fixed times, resulting in periodic patterns of the traffic generated [37]. On the other hand, Slowloris works by establishing long-lived sessions at the application layer, resulting in characteristic connection times [16]. However, the use of temporal features is not optimized in the literature:

- Early research (for example, Singh [37]) uses analytical as well as simulation approaches rather than a detection-focused approach;
- ML research (for instance, Reed [29], Jha [16]) occasionally recognizes the significance of time but fails to identify and contrast time-related characteristics in classifiers;
- Hybrid models (Karakaya [18], Khaldy [21]) have potential, though they are not designed specifically for the unique features of RoQ and Slowloris attacks.

Based on the analysis, the following can be concluded, the literature review effectively points out the problem but does not present a full, repeatable, and comparative solution that encompasses the following:

1. Generation of RoQ traffic, which is reproducible and cannot be generated by available public tools;
2. Systematic temporal feature engineering;
3. Comparison among several classifiers rather than one or two;
4. A hybrid approach that integrates fuzzy logic, ML, and distance metrics.

2.6.3 Critical Limitations Blocking Progress And Why This Dissertation Addresses a Real Gap

Currently, four significant limitations impede the progress of the identification of stealthy DoS attacks such as RoQ and Slowloris. Each limitation is a specific research gap that was targeted by this dissertation:

2.6.3.1 Limitation 1: Reproducibility Crisis → Solution 1: Reproducible RoQ Traffic Generator

The Problem: Currently, there is no publicly available tool capable of producing traffic for RoQ. As such, simulation or use of a proprietary tool was required, which makes it impossible to reproduce results.

The Solution: Based on the TCP rate adaptation model, a reproducible RoQ traffic generator was proposed to overcome the difficulties in making the traffic reproducible.

2.6.3.2 Limitation 2: Underemphasis on Temporal Features → Solution 2: Temporal Feature Extraction and Accuracy of Detection

The Problem: Even as the oscillatory pattern represents the hallmark of RoQ and the duration-of-connection pattern represents the hallmark of Slowloris, the literature on ML techniques gave greater importance to statistical features (packet rate, byte amount) than temporal features (inter-arrival entropy, burst periodicity, duration oscillations).

The Solution: Features such as burst periodicity, inter-arrival entropy, and duration entropy were carefully identified and assessed along with traditional features. This helped overcome the basic misalignment between the nature of attacks and the design of detection mechanisms.

2.6.3.3 Limitation 3: Lack of Systematic Comparison Studies → Solution 3: Systematic Classifier Benchmarking

The Problem: The literature tends to assess the performance of a single classifier or make comparisons with the performance of the baselines. A proper comparison of the performance of the existing classifiers, like RF, SVM, MLP, LightGBM, AdaBoost, and KNN, with RoQ/Slowloris on the same dataset with the same features hasn't been done yet.

The Solution: A set of eight or more classifiers (KNN, MLP, SVM, RF, LightGBM, AdaBoost, Naive Bayes, and the baselines) was tested on the same controlled data set, and this represented the first systematic benchmark for the detection of RoQ/Slowloris attacks.

2.6.3.4 Limitation 4: Sparse Hybrid Classification → Solution 4: Hybrid Model Specialization and Robustness

The Problem: Even though hybrids are theoretically beneficial, they are hardly used in RoQ/Slowloris, and if used, their results were not analyzed in a way to justify their

complexity over other ML classifiers.

The Solution: With the combination of fuzzy logic, the best-performing classifier, and distances adapted to the characteristics of the RoQ/Slowloris attack (fuzzy rules for oscillations and Euclidean distances for timing similarities), it was possible to provide proof of concept that hybrid models can be more effective in the attack domain than models based solely on ML.

2.6.4 Summary of Contributions and Expected Impact

Contributions to the Field from this Dissertation:

- **For the research community:** An easily reproducible methodology with open-source tools for the generation of RoQ traffic, which enable other studies to start from an equitable baseline;
- **For Practitioners:** Systematic benchmark study specifying the classifiers that perform best for the detection of RoQ/Slowloris attacks and under which circumstances the hybrid classifiers are justified by the increased complexity;
- **For the security industry:** Developing a hybrid detection framework that provides improved robustness in a realistic and noisy environment, and which guides IDPS integration.

3

Detection of Slowloris Attacks

3.1 Introduction

This chapter provides a system for the detection of Slowloris attacks. Slowloris is an intelligent, application-level denial of service attack that exhausts a servers connection pool through its long opening of HTTP connections [11]. The chapter is organized like this:

- Section 3.2 describes how Slowloris attacks can be detected specifically;
- Section 3.3 presents the proposed detection model and test environment;
- Section 3.4 defines the classification functions and ML algorithm used;
- Section 3.5 demonstrates and interprets the obtained results in two datasets for evaluation;
- Section 3.6 draws conclusions based on the experiments.

3.2 Use of Detection Methods for Slowloris Attacks

There are many methods to detect Slowloris and other slow HTTP attacks, and these methods differ in their effectiveness.

3.2.1 Approach with Statistical Methods

Statistical methods monitor network connection parameters, including, but not limited to the number of concurrent semi-open connections, average request completion time, and header arrival rate. Thresholding variants have two versions such as, static and adaptive. Static versions set a threshold, then send an alert when these metrics exceed this value [12]. The major weakness is the fragility of this threshold, that is either too low to prevent many false positives, or too high to prevent attacks. Adaptive statistical methods such as those based on exponential smoothing or entropy analysis provide more robust means by dynamically adapting threshold values based on baseline traffic parameters [2]. Even these methods are susceptible to attacks designed to stay under each detection threshold.

3.2.2 Machine Learning Approaches

Supervised ML algorithms have also demonstrated very promising performance when identifying slow lorises with sufficiently large datasets. The Random Forest, Gradient Boosting and SVM algorithms can take advantage of the temporal and volume characteristics of slow connections, so these models have performed well in experiments [5, 1]. These models require representative training data, and are also vulnerable to the differences in the distribution between training environment and prediction environment, thus should be tested on real-world dataset which is out of test environment.

3.2.3 Deep Learning Approaches

The temporal persistence exhibited by Slowloris streams, differentiating them from standard, sustainable connections, was modeled using recurring neural networks and LSTM structures [11, 30]. Unsupervised, autoencoders, trained solely on legitimate traffic, present an unsupervised approach and detecting abnormally high reconstruction error without knowledge of whether the traffic is malicious. However, deep learning techniques are more computationally expensive, and significantly more susceptible to hyperparameter choices, than traditional ML techniques.

3.2.4 Hybrid Approaches

These hybrids combine simple statistical models with a ML classifier or embed a fuzzy logic module to overcome the uncertainty introduced in the marginal data streams [18, 9]. They dynamically alter their detection thresholds, thus eliminating false positives, in scenarios where the legitimate traffic varies significantly over time. These represent the best performing techniques so far to detect undetectable DoS attacks, but require that each module be tuned accurately.

3.2.5 Datasets

Finding a suitable dataset has always been an issue for researching Slowloris. The majority of publicly available benchmarks such as KDD Cup 99, NSL-KDD or CICIDS2017 does not contain any application layer attack traffic with slow HTTP connection to be directly compared against [26, 32, 33]. Two datasets were used in this work:

- **Dataset A - Laboratory Dataset (`dataset_slowloris.csv`):** This dataset was created in a laboratory testbed setup that is detailed in Section 3.3.2. `Tshark` was used to capture the traffic of parameterized Slowloris attack on an Apache web server with harmless HTTP traffic also flowing through the setup at the same time. It contains 2,773 flows (bidirectional) of which, 1,768 (63.8%) attacking and 1,005 (36.2%) normal;
- **Dataset B - IFTO Dataset:** Publicly available LAN dataset, generated from the internal network infrastructure of the Swiss Federal Institute of Education, Science

and Technology of Tocantins (IFTO) and reported by Vinicius et al. [31] consists of 18,860 data flows describing actual network traffic under conditions of the Slowloris attack, with 8,777 attack flows (46.5%) and 10,083 legitimate data flows (53.5%). The almost balanced classes and diversity of legitimate traffic makes the dataset relevant for demonstrating generalization not in controlled lab environment.

Two datasets were used, generated under entirely separate circumstances. Dataset A tests the recognition of objects under completely controlled, replicable conditions, while Dataset B tests generalization of trained models to situations other than controlled lab environments.

3.3 Proposed Detection Model

3.3.1 Introduction to the Proposed Detection Model

The proposed structure was founded in an offline classification stream-based paradigm: raw data packets (`.pcap`) are inspected by using `tshark`, in order to extract statistics and time-based features for every stream. These values are used as input to a set of supervised and unsupervised learning models. The pipeline works in post-processing fashion allowing the reproducible benchmark and comparison between different datasets.

The evaluation approach uses a double-dataset approach, with training and validation performed on Dataset A. For evaluation within distribution, 5-fold stratified cross-validation is performed to determine a model's accuracy and effectiveness. Then, accuracy is calculated for model against dataset B to check its performance on unseen data outside of distribution. With this approach, specifically checks to see if the features selected to classify the traffic from within a carefully constructed lab traffic scenario can generalize to a realistic institutional network environment.

3.3.2 Experimental Testbed Environment

The lab environment was configured using an OVHCloud setup comprising four instances acting as:

- The **Victim** (`IP → 51.255.166.85`): It was targeting a web server running Apache 2.4;
- The **Attacker** (`IP → 51.255.166.133`): It used the Slowloris tool and kept a user configurable amount of slow http requests to the target server;
- The **Benign Traffic Generator** (`IP → 51.255.166.105`): It generated normal HTTP traffic to the target server;
- The **Monitor** (`IP → 51.255.166.114`): It passively listened to all network traffic using `tshark` on a promiscuous mode and stored network data to a `.pcap` file.

A standard procedure was followed throughout: first, HTTP traffic was sent normally over the network as a reference during the "reference period" and then the experiment entered into a "Slowloris attack" period, the whole session was captured by `tshark`. Later feature extraction was done offline from captured data.

3.3.3 Testbed Components

- **Traffic Generation**: Benign traffic was generated, it simulated legitimate HTTP GET requests to the target by means of the `curl` command and `Apache Benchmark (ab)`. Slowloris traffic was generated using *slowhttptest* on the attacker's instance. This tool was configured to open and maintain a large number of concurrent, incomplete HTTP connections to the victim's Apache server, periodically sending incomplete headers to keep each connection alive. The attack was run for a predetermined duration to ensure a representative traffic volume;

- **Feature Extraction Pipeline**: All the captured packets were analyzed with `tshark` and array analyzers to pull out the records for each data stream. Arrival time characteristics were extracted (`iat_mean`, `iat_std`) and found by calculation based on the packet capture, that was `frame.time_delta_displayed`. The `frame.time_delta_displayed` displays the real arrival time difference of each successive packet and, unlike a simple ratio, retains the irregularity characteristic to Slowloris;

- **Pre-processing**: All the features were scaled with `StandardScaler` (zero mean, unit variance) before being applied on the models. A binary classification `label` $in\{0$ (benign), 1 (attack)$\}$ was employed as target for the supervised learning models.

3.4 Specific Detection Components and Algorithms

3.4.1 Classification Features

The choice of flow-level features was based on their ability to represent behavior related to Slowloris type traffic. Long duration of flow, very low packet rate, very small packet size (because HTTP headers are incomplete) and variability of inter-arrival times are features which characterize Slowloris attacks. Table 3.1 shows the comparison of relevancy among 9 extracted features for each dataset.

Feature	Description	Dataset A	Dataset B
Duration	Flow duration (s), from first to last packet	Moderate	Very High
pkts_total	Total number of packets in the flow	Very Low	Low
bytes_total	Total bytes transferred	High	Moderate
pkt_size_mean	Mean packet size (bytes)	High	High
tcp_bytes_total	Total TCP payload bytes	Very High	Moderate
tcp_pkts	Number of packets with TCP payload > 0	Very Low	Very Low
pkts_rate	Packet rate (pkts/s = pkts_total / duration)	Very High	Very High
iat_mean	Mean inter-arrival time between packets	Moderate	High
iat_std	Standard deviation of inter-arrival times	Moderate	Moderate

Table 3.1: Flow-level features used for Slowloris detection: comparison between Dataset A and Dataset B.

SHAP value-based feature importance analysis on the random forest model (Dataset A) shows that the two optimal discriminators are `pkts_rate` (|SHAP| mean = 0.1575) and `duration` (0.1525), and on dataset B, the two optimal discriminators are `duration` (|SHAP| mean = 0.3344) and `iat_mean` (0.2204), which is consistent with the feature design.

3.4.2 Detection with Supervised ML

Eight supervised classifiers were tested and were chosen so as to represent the important families of ML algorithms and enable a proper comparison:

- **KNN**: Instance-based baseline; feature scaling sensitive and shows the preprocessing pipeline works;
- **SVM**: Maximum-margin classifier with RBF kernel; perform well with high dimensional feature spaces;
- **RF**: Ensemble decision tree clustering; provide feature importance estimates by minimizing average impurity;
- **MLP**: Feed-forward neural network; test whether or not deeper models give advantages compared with the classical ones.
- **GB / LightGBM / AdaBoost**: Family of features supported; allows the testing of different ensemble strategies and gradient boosting implementations;

- **Naive Bayes**: Probabilistic baseline; act as a lower bound and shows the importance of the independence assumptions of the features.

All models supervised were trained using a stratified 5-fold cross-validation ensuring that the proportion between the classes were maintained throughout the convolution phase. Accuracy, precision, recall, F1-Score, ROC-AUC, False Positive Rate (FPR) and False Negative Rate (FNR) were the six measures used for performance evaluation. In order to simplify analysis, the measures FPR and FNR are of importance since they directly correspond to actual operational characteristics when dealing with detecting attacks. FPR measures false alarms that waste security analysts time and effort while FNR measures undetected attacks that have the potential to drain resources at the server.

Significance testing of differences in pairwise performances was performed using McNemar's test [8], which is a non-parametric test that assess whether two classifiers have different error tendencies using differing predictions placed in a contingency table. The significance level was established at $\alpha = 0.05$.

3.4.3 Detection with Unsupervised ML

The three unsupervised methods were applied and trained only on normal traffic as to simulate a deployment where no attack label is present:

- **K-Means Clustering**: Two configurations were chosen to study: $k = 2$ (with the assumption of a natural binary class nature), and the k value that leads to the minimal Davies-Bouldin index on the validation set ($k = 7$ for both datasets A and B). Data streams were allocated to cluster by majority voting and cluster that the centroid distance from the normal training distribution is large were declared to be an attack clusters;

- **Isolation Forest**: Anomaly detection by random recursive partitioning. Contamination parameter is needed, a search was performed using the values $\in \{0.05, 0.10, 0.20, 0.30, 0.40, 0.50\}$, and the value 0.05 was selected because it performed the best (high accuracy, acceptable recall). It is represented as a mean ± standard deviation over 10 random initial values;

- **Autoencoder**: When a deep autoencoder is trained on the normal traffic flows, it reconstructs the normal traffic flows. The difference of normal traffic flow and reconstructed normal traffic flow is the anomaly score. The detection threshold was set at p90 (90% percentile) of the reconstruction error on the normal validation data. The choice of the architecture is based on comparison of a shallow architecture (one hidden layer) and a deep architecture (three hidden layers). It turned out that the deep architecture worked better. The data shown are means ± standard deviation, which were averaged over 10 random initialization in order to deal with training instability.

3.5 Observed Results

3.5.1 Observed Metrics

The most important metrics for evaluation are the F1-Score, FPR, FNR and ROC-AUC. Accuracy is reported for completeness, but is less significant due to the varying size of classes in Dataset A. The quality of clustering of the unsupervised models is also reported, through the Silhouette Score and Davies-Bouldin Index of the K-means algorithm.

For the stochastic models (Isolation Forest and Autoencoder), the values are presentedas a mean ± standard deviation out of 10 different initializations. For the supervised models with cross validation, the standard deviation is calculated of the F1-Score of the different folds.

3.5.2 Detection Results with Supervised ML

In table 3.2 the supervised ML outputs obtained on Dataset A (`dataset_slowloris.csv`) by applying the 5-Fold Stratified Cross-Validation approach are shown. In table 3.3 the results on Dataset B are shown.

Model	Accuracy	Precision	Recall	F1	AUC	FPR	FNR
KNN	0.9996	1.0000	0.9994	0.9997	0.9997	0.000	0.001
SVM	0.9996	1.0000	0.9994	0.9997	1.0000	0.000	0.001
RF	1.0000	1.0000	1.0000	1.0000	1.0000	0.000	0.000
MLP	1.0000	1.0000	1.0000	1.0000	1.0000	0.000	0.000
GB	1.0000	1.0000	1.0000	1.0000	1.0000	0.000	0.000
LightGBM	1.0000	1.0000	1.0000	1.0000	1.0000	0.000	0.000
AdaBoost	1.0000	1.0000	1.0000	1.0000	1.0000	0.000	0.000
Naive Bayes	0.9978	0.9966	1.0000	0.9983	0.9975	0.006	0.000

Table 3.2: Supervised ML detection results (5-Fold Stratified CV) on Dataset A.

Model	Accuracy	Precision	Recall	F1	AUC	FPR	FNR
KNN	0.9777	0.9811	0.9708	0.9759	0.9865	0.0163	0.0292
SVM	0.9453	0.9337	0.9499	0.9417	0.9892	0.0587	0.0501
RF	0.9800	0.9755	0.9819	0.9786	0.9991	0.0216	0.0181
MLP	0.9722	0.9657	0.9756	0.9703	0.9982	0.0307	0.0244
GB	0.9799	0.9710	0.9866	0.9785	0.9992	0.0260	0.0134
LightGBM	0.9796	0.9756	0.9810	0.9782	0.9991	0.0215	0.0190
AdaBoost	0.9791	0.9816	0.9732	0.9774	0.9991	0.0159	0.0268
Naive Bayes	0.8856	0.8114	0.9826	0.8888	0.9657	0.1988	0.0174

Table 3.3: Supervised ML detection results (5-Fold Stratified CV) on Dataset B.

The treated ideal results of all five models, such as RF, MLP, GB, LightGBM, and

AdaBoost, was a function of the structured setting of Dataset A. The uniform traffic-generation process produces a feature space that is much more easily distinguished than the heterogeneous traffic of Dataset B, which is the true unsystematic case.

On Dataset B, all models are still competitive but performance is noticeably degraded compared to Dataset A, reflecting the distribution shift due to real-world institutional heterogeneous traffic. RF and GB provide a good trade-off between metrics (F1-Score = 0.9786 and 0.9785 respectively) and RF has the lowest FNR (0.0181) for any tree-based classifier. This is operationally significant, as missed attacks can result in persistent server exhaustion. AdaBoost provides the lowest FPR (0.0159) at the expense of a slightly higher FNR (0.0268). NB shows a large degradation on Dataset B (FPR = 0.1988), which indicates the violation of the feature-independence assumption for this dataset where temporal features are heavily correlated.

There is also an exceptionally large training time reported for LightGBM on Dataset A (68.9s) owing to the cold start time of the library during the first execution. After which subsequent runs show the normal training time is around the same as RF (1.5s) and is still the fastest at inference time (0.8 ms).

3.5.2.1 Statistical Significance - McNemar Test

The McNemar test was applied on Dataset B to determine if differences between pairwise classifier performances are significant. Selected pair results are shown in Table 3.4.

Pair	b	c	p**-value**	**Significant?**
RF vs Naive Bayes	1942	161	< 0.001	Yes
RF vs KNN	137	94	0.0056	Yes
RF vs SVM	861	206	< 0.001	Yes
SVM vs Naive Bayes	1578	452	< 0.001	Yes
KNN vs Naive Bayes	2001	263	< 0.001	Yes

Table 3.4: McNemar test results for selected classifier pairs on Dataset B (IFTO). b: errors by model A only; c: errors by model B only; significant at $\alpha = 0.05$.

All pairs tested showed a statistically significant difference in performance ($p < 0.05$). The comparison between RF and Naive Bayes produced $b = 1942$ and $c = 161$, showing that the errors made by the two classifiers had a very different distribution. This result was highly significant and suggests that RF has a more trustworthy error distribution on Dataset B. Even the comparison between RF and KNN, with a p-value of 0.0056, was significant. This small difference of 0.0027 in F1-Score, if not due to chance, must represent a real advantage of RF on this dataset.

3.5.3 Detection Results with Unsupervised ML

Table 3.5 and Table 3.6 show the unsupervised ML results in Dataset A and Dataset B respectively. The unsupervised models were trained using only benign flows.

Model	Type	Precision	Recall	F1	AUC	FPR	FNR
K-Means (k=2)	Clustering	0.7933	1.0000	0.8848	1.0000	0.500	0.000
K-Means (k=7)	Clustering	0.9995	1.0000	0.9997	1.0000	0.001	0.000
Isolation Forest	Anomaly Detection	0.9745	1.0000	0.9871	0.9982	0.0502	0.000
Autoencoder	Anomaly Detection	0.9926	1.0000	0.9963	1.0000	0.0144	0.000

Table 3.5: Unsupervised ML detection results on Dataset A (Isolation Forest and Autoencoder report mean ± std over 10 seeds).

Model	Type	Precision	Recall	F1	AUC	FPR	FNR
K-Means (k=2)	Clustering	0.5452	0.0199	0.0385	0.9480	0.0145	0.9801
K-Means (k=7)	Clustering	0.9385	0.1479	0.2555	0.7980	0.0084	0.8521
Isolation Forest	Anomaly Detection	0.7757	0.1989	0.4233	0.9337	0.0498	0.7103
Autoencoder	Anomaly Detection	0.8780	0.8346	0.8511	0.9299	0.0984	0.1654

Table 3.6: Unsupervised ML detection results on Dataset B (IFTO, n = 18,860). Isolation Forest and Autoencoder report mean ± std over 10 seeds.

Dataset A reflects that the methods clearly separated benign from attack traffic. While there is no prior knowledge regarding the optimal cluster quantity k, K-Means with $k = 7$ obtained optimal results, precision of 0.9995, recall of 1.0000, F1-Score of 0.9997, ROC-AUC of 1.0000, and FPR of 0.001. K-means with $k = 2$ demonstrated subpar performance, as indicated by its FPR of 0.5, the appropriate number of clusters on this dataset is essential. Isolation Forest also fared decently by achieving precision of 0.9745, recall of 1.0, F1-Score of 0.9871, ROC-AUC of 0.9982 and FPR of 0.0502. The autoencoder did the best for balancing the anomaly detection, that achieved precision of 0.9926, recall of 1.0, F1-Score of 0.9963, ROC-AUC of 1.0 and FPR of 0.0144. This demonstrated that Slowloris was easily learnable when learning from normal traffic.

With the real-world traffic of Dataset B, the unsupervised approach begins to struggle compared with that for Dataset A. Both configurations of the K-Means algorithm demonstrate poor recall for IFTO traffic, FNR of 0.9801 and 0.8521 for $k = 2$ and $k = 7$ respectively, suggesting that institutional traffic Slowloris flows are not tightly grouped and distinct within this feature space. The Isolation Forest demonstrates equally poor performance with `contamination = 0.05`, the FNR is 0.7103. Improving this recall requires increasing contamination, thus lowering precision as demonstrated by the contamination sweep (at `contamination = 0.40` the FNR is 0.000, but the precision is only 0.685). Despite the seed-to-seed variation in its performance (F1-Score 0.718-0.920), the autoencoder proves to be the most effective model when used for the unsupervised approach on Dataset B, achieving an average F1-Score of 0.8511 ± 0.088. It is advised that the median score across all random initializations is used as the threshold, to improve reliability.

3.5.3.1 Isolation Forest Contamination Sweep

The contamination parameter indicates how many training data points are assumed to be outliers. Figure 3.1 displays the sweep result on Dataset B. Using contamination 0.10, the isolation forest performs with F1-Score 0.9083 (precision = 0.8897, recall = 0.9277, FNR=0.0723). This again demonstrates that contamination parameter is crucial to the fast calculation. Nevertheless, its usefulness in real applications is limited as one has to know the attack frequencies.

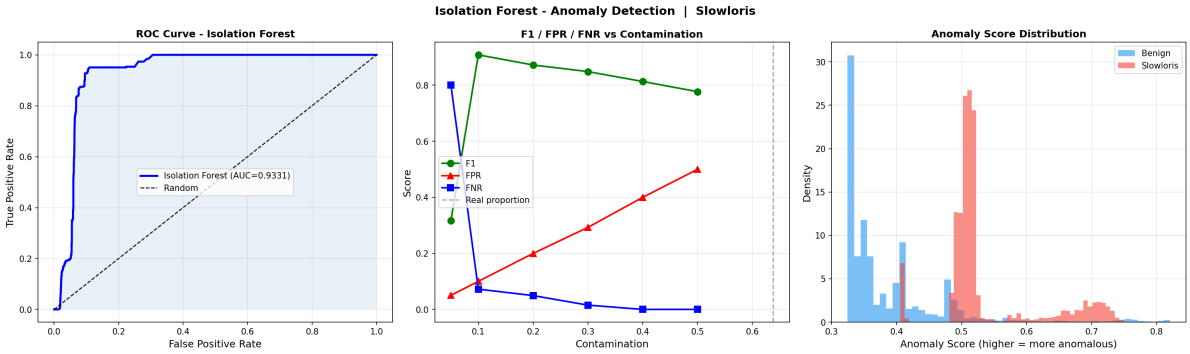


Figure 3.1: Isolation Forest performance sweep across contamination values on Dataset B.

3.5.3.2 Autoencoder Threshold Sensitivity

The threshold p90 was chosen by sweeping through reconstruction error percentiles for the benign validation set and finding the maximum F1-Score. From the threshold sensitivity plot (Figure 3.2) it can be seen that p90 seems to be a reliable point for operation and is a point where is maintained a balance between FPR and FNR given the traffic distribution.

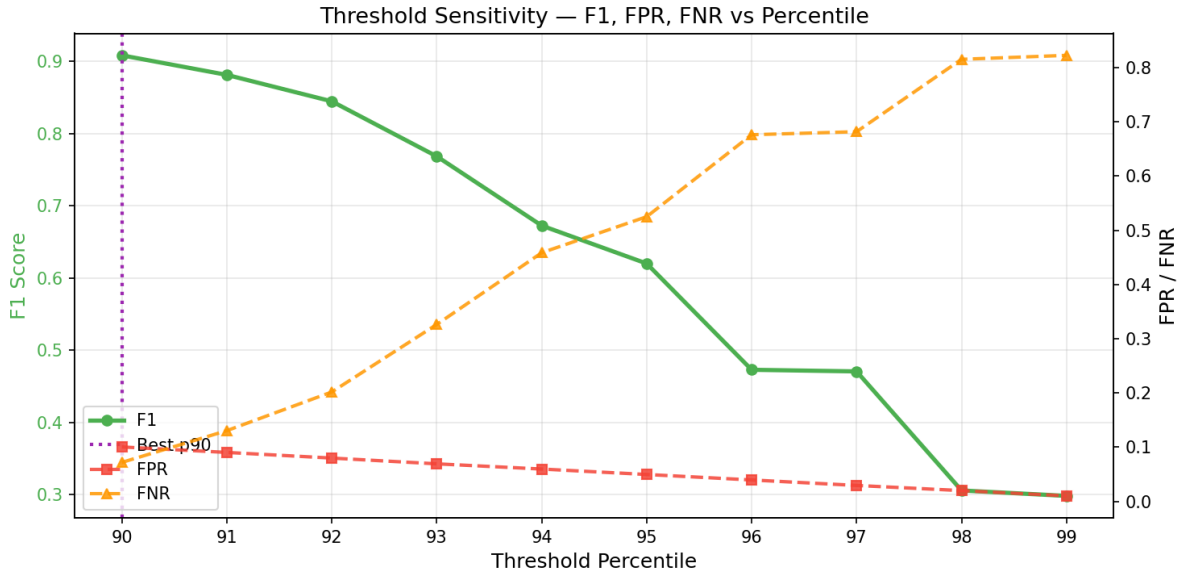


Figure 3.2: Autoencoder threshold sensitivity on Dataset B.

3.5.4 Summary of Observed Results

An across-dataset summary of the best performance per category is presented in Table 3.7.

Category	Model	Dataset A (Lab)			Dataset B (IFTO)		
		F1	FPR	FNR	F1	FPR	FNR
Supervised (best F1)	RF	1.0000	0.000	0.000	0.9786	0.0216	0.0181
Supervised (low FPR)	AdaBoost	1.0000	0.000	0.000	0.9774	0.0159	0.0268
Supervised (low FNR)	GB	1.0000	0.000	0.000	0.9785	0.0260	0.0134
Unsupervised (best F1)	K-Means (k=7) / Autoencoder	0.9997	0.001	0.000	0.8511	0.0984	0.1654

Table 3.7: Best-performing model per category across both evaluation datasets.

In both datasets supervised methods do better than unsupervised methods. Supervised models are only marginally degraded from Dataset A to B, by 0.02 F1-Score on RF which suggests that the flow-level temporal features derived from lab traffic captured patterns of behavior generalisable to the live institution network. The difference is significantly greater for unsupervised models like K-means and Isolation Forest, which suggests that unsupervised detection of anomalies via clustering shape is highly dependent on the distribution characteristics of the target environment. The most transferable of the unsupervised methods is autoencoder, which holds F1-Score = 0.851 on IFTO without any attack labels during training.

3.6 Conclusions

In this chapter, a flow level Slowloris detection framework utilizing supervised and unsupervised ML algorithms has been designed, implemented, and analyzed using two datasets of completely different nature.

The models that are suitable to be used in practice are Random Forest and Gradient Boosting. The two models that achieved the optimal FNR and FPR trade-off on the IFTO dataset (RF F1-Score = 0.9786, FNR = 0.0181; GB F1-Score = 0.9785, FNR = 0.0134) were also shown, using the McNemar test, to achieve results which are statistically significant when compared to inferior baselines (NB and SVM). The near perfect results on Dataset A for the eight tree-based and neural models indicate that the flow features can cleanly distinguish between Slowloris and benign traffic in a laboratory setting. In the IFTO dataset, the majority of these features have been shown to carry over to a realistic institutional environment.

The K-Means ($k = 7$) and autoencoder (deep one) were the most performing of the unsupervised models for the Dataset A and Dataset B, respectively. The results in term of F1-Score on the Dataset A are F1-Score = 0.9997 and on the Dataset B F1-Score = 0.851. The major drawbacks is the lack of training stability depending on the seed of the learning which require the usage of multi-seed approach and selection of the median performing model for the actual system. K-Means and Isolation Forest work well on the lab traffic, but on the IFTO traffic, performance drops, so these methods can't be used as only detection mechanism.

Feature importance using SHAP indicates that the temporal features, specifically `duration`, `pkts_rate`, `pkts_total` and `tcp_bytes_total` are the best predictors of Slowloris and justify the choices of feature engineering and support interpretability of the output alerts of the detection framework.

The main identified weaknesses are:

1. The lab-generated dataset uses only one andhomogeneousattack configuration;
2. The contamination parameter of Isolation Forest is not known a-priori and it depends on the occurrence of attack;

4

Detection of RoQ Attacks

4.1 Introduction

This chapter provides a detection framework for RoQ attacks. RoQ attacks exploit the congestion control mechanisms of TCP to periodically reduce the throughput of legitimate flows, instead of producing very large volumes of traffic as with volumetric DoS attacks. These characteristics of RoQ attacks make them difficult to detect with signature or threshold based approaches and hence ML based approaches are proposed for detecting them.

The organization of the chapter is given below:

- The different detection mechanisms to detect RoQ attacks are detailed in Section 4.2;
- The suggested detection model and the experimental testbed are given in Section 4.3;
- The classification features and ML algorithms are listed in Section 4.4;
- The results are shown and explained in Section 4.5;
- In Section 4.6 conclusions from the experiments are given.

4.2 Use of Detection Methods for RoQ Attacks

These RoQ attacks are designed such that when large bursts of traffic are sent at the timescale of TCP congestion control, the congestion window of the server fluctuates significantly. In comparison to Slowloris which works at the level of a connection in the application layer, the transport layer attacks carried out by RoQ give rise to a distinctive pattern of oscillation in throughput, inter-arrival times of packets and pattern of burst structure. These features may help to detect these RoQ attacks through varying approaches.

4.2.1 Approach with Statistical Methods

Statistical methods to estimate packet rate, inter-arrival time variance, and burst periodicity can detect the abnormal oscillation of RoQ. Using adaptive thresholds on the burst

parameters could detect the periodic collapse of the TCP congestion-window which identifies RoQ. The false positive rates of fixed thresholds are normally high due to the fact that RoQ can behave similarly to legitimately bursty traffic, especially in heterogeneous networks.

4.2.2 Machine Learning Approaches

Supervised ML classifiers excel at modeling non-linear and complicated relationships that exist between the burst and size of packets to distinguish between RoQ and normal traffic. RF, GB and SVM all have shown an aptitude for learning discriminating boundaries with flow-level features like volume, burst metrics, and TCP metrics. The problem with RoQ is the lack of available data sets forcing papers to use artificial traces with few varieties.

4.2.3 Deep Learning Approaches

By using only normal traffic to train autoencoders, they could have built a compact model that represents normal flow traffic, so that, an anomaly, such as RoQ attack flows are easily identified as anomalies because, deviation from the normal flow pattern (that RoQ flows due to abnormal bursts do), produce large reconstruction errors, when trained on a normality model. Recurring structures like LSTMs could have adapted to detect the temporal oscillations in the RoQ traffic, but their computational cost would have restricted their usability.

4.2.4 Hybrid Approaches

Hybrid models usually mix statistical methods with ML and sometimes, distance based or uncertainty aware components. Such a modeling seems particularly interesting for RoQ attacks as their malicious activity can be spotted as a set of small oscillations rather than one outstanding signature or feature. In this case, hybrid architecture architectures can be valuable in bridging distinct approaches that reveal complementary view of the target traffic signature.

However, these models usually imply more effort to calibrate and can challenge the readability regarding the impact of specific variables in the decisions.

Henceforth, this chapter aims at evaluating some of simple unsupervised and supervised approaches before making conclusion about the optimal pathway for an attack classification.

4.2.5 Dataset

Lack of accessible RoQ traffic dataset is already established in literature, where no existing public benchmark dataset was found that consists of labeled RoQ flow and enough temporal diversity. Hence, a new RoQ traffic dataset was created within a laboratory testbed:

- **RoQ Laboratory Dataset (`dataset_roq_window_full.csv`)**: Generated using custom made RoQ traffic generator, using the TCP rate-adaptation model within the testbed outlined in Section 4.3.2. Consists of 788 window-level flows consisting of 180 attack flows (22.0%) and 608 benign flows (78.0%), recorded under parameterized RoQ attacks of varied burst size and inter-burst time. Unlike the case study in Chapter 3, where the availability of an external validation set, all evaluations were made using 5-fold stratified cross validation on the testbed described herein.

4.3 Proposed Detection Model

4.3.1 Introduction to the Proposed Detection Model

The proposed detection scheme followed the same offline, flow-level classification pipeline described in Chapter 3. Packet captures were first parsed (`.pcap`) using `tshark` to extract statistical and time features for each time window. These feature vectors are then fed to a battery of supervised and unsupervised models. The pipeline ran in post-processing mode thus facilitating reproducible comparison. As opposed to the Slowloris pipeline, this scheme aggregated per-flow statistics by time windows rather than per-flow. As RoQ attacks produce an oscillation over multiple connections through time, features were extracted that capture bursty pattern of RoQ attacks using a fixed-duration moving window.

The validation used 5-fold stratified cross validation on the RoQ laboratory data set which maintains the class ratios. Reports for the accuracy, precision, recall, F1-Score, ROC-AUC, FPR, and FNR was the mean standard deviation on 5-folds.

4.3.2 Experimental Testbed Environment

The experimental setup was hosted on OVHCloud servers in the form of different entities to generate the benign traffic, to carry out the attacks and to capture the traffic. The captured traffic was converted into flow level and fed into the ML models.

The whole process generally took the following path:

1. Generation of the normal HTTP traffic as the base;
2. Transition to RoQ attack state;
3. Capture and offline processing of the whole experiment run.

4.3.3 Testbed Components

- **Traffic Generation**: Benign traffic was produced by launching real HTTP GET requests to the target server using both `curl` and `ab` (Apache Benchmark). The RoQ generator, implemented in Python using `asyncio` and `aiohttp`, operates in a square wave manner (as proposed by Guirguis et al. [13]). It sends δ *requests*/s for τ seconds

(the *burst* period) and then stops sending requests for $T - \tau$ seconds (the *idle* period). The average request rate of the generator is given by,

$$\bar{r} = \frac{\delta \cdot \tau}{T}, \tag{4.1}$$

which is maintained to be small enough to remain *stealthy*. The `force_close = True` parameter on the connector ensures that every *request* closes the underlying TCP connection after sending, thus maximally penalizing the server's *congestion window*. The generator's parameters (T, τ, δ, number of cycles) are all settable through the command line, as shown in Table 4.1;

- **Feature Extraction Pipeline**: The captured packet files were parsed using `tshark`. From the per-packet timestamps, byte count and TCP flags (ACK, PSH etc) the values for the window-level features are calculated. For this task the script `aggregate_roq_window.py` was used which groups traffic into fixed time windows and then produces the feature set listed in Section 4.3.1;

- **Pre-processing**: All features were zero-mean unit-variance scaled (using `StandardScaler`) before being input to the models. A binary classification label 0, 1 is used, where 0 means benign traffic and 1 means attack traffic as the target variable for supervised learning models.

Parameter	Flag CLI	E1	E2	E3	E4	Description
T	`-period`	10 s	10 s	20 s	10 s	Total period of the cycle
τ	`-burst-duration`	1,5 s	1,5 s	2,0 s	0,8 s	Duration of the *burst*
δ	`-burst-rate`	150 req/s	300 req/s	200 req/s	400 req/s	Rate during the *burst*
n	`-cycles`	30	30	15	30	Number of cycles

Table 4.1: Configurable parameters of the RoQ generator per experiment.

4.4 Specific Detection Components and Algorithms

4.4.1 Classification Features

The selection of features in the window was driven by features found in RoQ traffic, such as anomalous packet size distribution, large fluctuation in burst intensity, increased TCP ratio compared to benign traffic and peculiar periodicities in the arrival interval time series. Table 4.2 summarizes the 14 features computed on each time window and their interest for RoQ identification.

Feature	Description	Relevance to RoQ
pkt_size_std	Standard deviation of packet sizes	High → RoQ bursts produce abnormal size variance
tcp_ratio	Fraction of packets carrying TCP payload	Very High → Attack elevates TCP payload ratio
pkt_size_mean	Mean packet size (bytes)	High → Burst packets tend to be larger than benign
burst_peak_ratio	Peak burst rate / mean rate	Very High → RoQ produces high peak-to-mean ratio
bytes_rate_cv	Coefficent of variation of byte rate	High → Rate oscillation is the hallmark of RoQ
burst_cv	Coefficient of variation of burst duration	Very High → Variable burst lengths indicate attack
tcp_bytes	Total TCP payload bytes in window	Moderate → Elevated during high-rate bursts
bytes_total	Total bytes transferred in window	Very Low → Higher during high-rate bursts
bytes_rate	Mean byte rate (bytes/s)	Low-Moderate → Inflated during burst phase
n_pkts	Number of packets in window	Very Low → Burst phase increases packet count
pkts_rate	Packet rate (pkts/s)	Very Low → Elevated during bursts
iat_cv	Coefficient of variation of inter-arrival time	Low-Moderate → Periodicity affects IAT distribution
iat_mean	Mean inter-arrival time between packets	Very Low → Less discriminative for RoQ than for Slowloris
iat_std	Standard deviation of inter-arrival time	Low-Moderate → Marginally captures burst irregularity

Table 4.2: Window-level features used for RoQ detection, with characterization relevance.

This shows how the three most discriminative features of the Random Forest model are confirmed with the SHAP-value-based feature importance analysis shown in Section 4.5.2 as `burst_cv` ($|\text{SHAP}|$ mean = 0.0788), `tcp_ratio` (0.0772) and `burst_peak_ratio` (0.0758) which correspond with how RoQ attacks works through manipulating the TCP-layer. The Mutual Information analysis does not align with the ranking based on one of the three most discriminative features of the Random Forest model. It shows the features in the order `burst_cv` (MI = 0.5215), `bytes_rate_cv` (0.5206) and `tcp_bytes` (0.5178).

4.4.2 Detection with Supervised ML

Eight supervised classifiers were chosen to sample across the major families of ML algorithms and provide a fair comparison:

- **KNN**: baseline instance-based, sensitive to feature scaling, used to indicate effectiveness of pre-processing pipeline;
- **SVM**: max-margin with RBF kernel, known to perform well in moderate-dimensional feature spaces;
- **RF**: ensemble of decision trees, capable of producing feature importance estimations, generally resistant to over-fitting;
- **MLP**: feed-forward neural network, testing if higher-level models offer benefit over traditional algorithms;
- **GB / LightGBM / AdaBoost**: family of gradient boosting models to compare different ensemble strategies and gradient boosting implementations;
- **Naive Bayes**: probabilistic baseline, provides a lower bound and illustrates the significance of independence assumptions.

All supervised models were trained by means of 5-fold stratified cross-validation, maintaining class distribution (22% attack, 78% benign) in all the folds. Seven metrics, accuracy, precision, recall, F1-Score, ROC-AUC, FPR and FNR, were used for evaluating performance, and are expressed as average standard deviation across folds.

4.4.3 Detection with Unsupervised ML

Three unsupervised techniques were applied and trained only on benign traffic data in order to simulate a deployment environment in which no attack label is available:

- **K-Means Clustering**: The number of clusters was optimized by the combination of the Elbow Method and silhouette analysis varying k over 2,... ,8. In both cases the minimum of Within-Cluster Sum of Squares (WCSS) and maximum of silhouette indicated $k = 3$. The data windows were assigned to clusters via majority voting. Those clusters which were further from the benign training distribution centroid, as indicated by distance between centroid and the mean of benign windows, were considered attack clusters;
- **Isolation Forest**: It is an unsupervised method of anomaly detection based on random recursive partitioning of data. A contamination parameter sweep was made over $values \in 0.05, 0.10, 0.15, 0.20, 0.25, 0.30, 0.35, 0.40, 0.45, 0.50$. The optimum contamination parameter of 0.32 was chosen in order to maximize the F1-Score on validation set. Results represent average standard deviation over 10 seeds;
- **Autoencoder**: A deep autoencoder trained only on benign traffic flow. The Mean Squared Error (MSE) over the reconstructed window is considered the anomaly score. Three architecture types were experimented, such as V1 (original) architecture, V2

(subset of top features) and V3 (Bottleneck + Dropout regularisation). Two strategies were used to determine the anomaly threshold:

1. A percentile optimizing F1-Score (max-F1);
2. A percentile insuring recall of 0.95 (min-recall 0.95).

Results represent average standard deviation over 10 seeds.

4.5 Observed Results

4.5.1 Observed Metrics

The main measures for evaluation are F1-Score, FPR, FNR and ROC-AUC. The accuracy is also given as a reference although not a very useful measure due to the data skew (78% benign, 22% attack). For supervised models, a standard deviation over the five folds for the F1-Score is reported due to using cross-validation to assess the stability. For the unsupervised models, the performance is reported for the chosen operating point. K-Means is also reported with the elbow and Silhouette criterion and autoencoder through reconstruction error distributions and with the threshold dependency.

4.5.2 Detection Results with Supervised ML

The ML supervised results generated using 5-fold stratified cross-validation on the RoQ lab dataset (`dataset_roq_window_full.csv`) is shown below in Table 4.3.

Model	Accuracy	Precision	Recall	F1	AUC	FPR	FNR
KNN	1.0000 ± 0.0000	1.0000 ± 0.0000	1.0000 ± 0.0000	1.0000 ± 0.0000	1.0000 ± 0.0000	0.0000 ± 0.0000	0.0000 ± 0.0000
SVM	1.0000 ± 0.0000	1.0000 ± 0.0000	1.0000 ± 0.0000	1.0000 ± 0.0000	1.0000 ± 0.0000	0.0000 ± 0.0000	0.0000 ± 0.0000
RF	0.9987 ± 0.0025	0.9946 ± 0.0108	1.0000 ± 0.0000	0.9973 ± 0.0055	1.0000 ± 0.0000	0.0016 ± 0.0033	0.0000 ± 0.0000
MLP	0.9987 ± 0.0025	0.9946 ± 0.0108	1.0000 ± 0.0000	0.9973 ± 0.0055	1.0000 ± 0.0000	0.0016 ± 0.0033	0.0000 ± 0.0000
AdaBoost	0.9987 ± 0.0025	0.9946 ± 0.0108	0.9973 ± 0.0055	0.9973 ± 0.0055	1.0000 ± 0.0000	0.0016 ± 0.0033	0.0000 ± 0.0000
LightGBM	0.9975 ± 0.0031	0.9892 ± 0.0132	1.0000 ± 0.0000	0.9945 ± 0.0067	0.9999 ± 0.0003	0.0033 ± 0.0040	0.0000 ± 0.0000
GB	0.9962 ± 0.0031	0.9892 ± 0.0132	0.9944 ± 0.0111	0.9917 ± 0.0068	0.9972 ± 0.0056	0.0033 ± 0.0040	0.0056 ± 0.0111
Naive Bayes	0.9873 ± 0.0090	0.9487 ± 0.0354	1.0000 ± 0.0000	0.9733 ± 0.0186	0.9902 ± 0.0068	0.0165 ± 0.0116	0.0000 ± 0.0000

Table 4.3: Supervised ML detection results (5-Fold Stratified CV) on RoQ Laboratory Dataset.

The KNN and SVM algorithm achieved perfect classification throughout the cross-validation folds, with all accuracy, precision, recall, F1-Score and ROC-AUC scores recorded 1.0000 with no FPR or FNR values noted. The following closest was the RF, MLP and AdaBoost, mean F1-Score recorded 0.9973 for both, and both recorded mean FPR (0.0016). The LightGBM recorded mean F1-Score (0.9945), and the GB had mean F1-Score of 0.9917. Naive Bayes mean F1-Score of 0.9733 is recorded the lowest of those assessed and also recorded the largest mean FPR of 0.0165.

Based on the above observations, the RoQ feature space seems very separable under the tested experimental conditions. The perfect performances obtained for both KNN and SVM have to be seen in context with the controlled experimental environment and also

considering the fact that cross-validation on the same data is done. From the above results it may be concluded that the features chosen carry high discrimination ability but do not solely guarantee a general solution.

As the F1-Score of these, in respect to best F1-Score, is very close to an optimal value, and the respective FPR is kept relatively low, RF, MLP and AdaBoost are very suitable alternatives. Naive Bayes results in zero FNR, while due to reduced precision and higher FPR it suggests that its underlying assumption of conditional independence is not valid for this feature-space in which the `burst`, `rate`, `packet_size` and `tcp_ratio` variables seem to be mutually dependent.

4.5.3 Feature Importance Analysis

Three diverse but complementary techniques were employed to quantify the feature importance like Random Forest Gini impurity, SHapley Additive exPlanations (SHAP) values, and Mutual Information (MI) as shown in Figure 4.1. The three methods were highly consistent regarding the rank of the most discriminatory features.

The order is not the same in all measure, but they all pinpoint a same group of variables. `burst_cv`, `bytes_rate_cv`, `burst_peak_ratio` and `tcp_ratio` remain key features on each technique. `iat_mean`, `bytes_total`, `pkts_rate` and `n_pkts` were relatively less importance on RF, SHAP importance, but still considered of moderate importance for MI. On the other hand, `pkt_size_mean`, `pkt_size_std`, and `tcp_bytes` consistently maintain a high degree of importance.

These outcomes are in line with the expected form of RoQ attacks. It's not simply the increase of the packet count and packet rate which reveals such an attack, rather RoQ is characterized by the variability of the bursts, by the variation of byte rate and by the correlation between the TCP payload and all the traffic. The decision made is indeed that we should not select a small set of features for the main models.

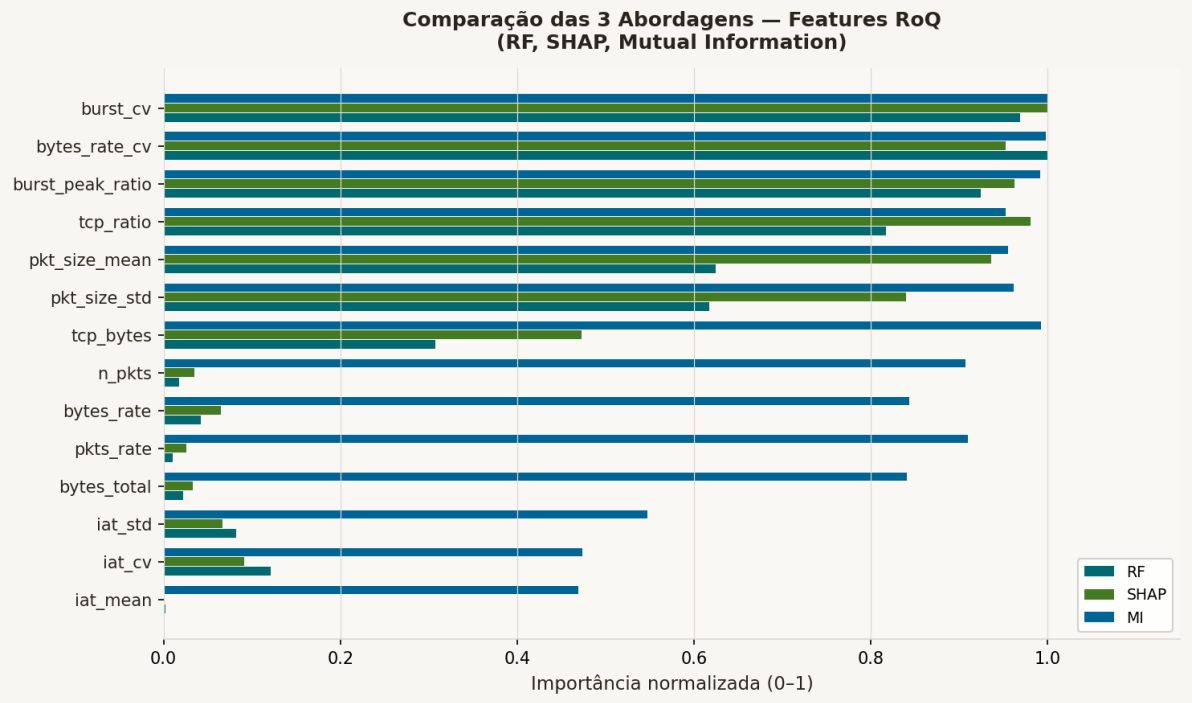


Figure 4.1: Feature importance comparison for RoQ detection.

4.5.4 Detection Results with Unsupervised ML

Benign flows were used to train the unsupervised models and the mixed attack and benign flow data was used for testing. The table bellow is the main result as reported Table 4.4.

Model	Accuracy	Precision	Recall	F1	AUC	FPR	FNR	TP	TN	FP	FN
K-Means	0.8287	1.0000	0.2500	0.4000	0.8950	0.0000	0.7500	45	608	0	135
Isolation Forest	0.9530	0.8488	0.9667	0.9039	0.9601	0.0510	0.0333	174	577	31	6
Autoencoder (V1 - Original)	0.9975	0.9890	1.0000	0.9945	0.9989	0.0033	0.0000	180	606	2	0

Table 4.4: Unsupervised ML detection results on RoQ Laboratory Dataset.

For the autoencoders with V1 - Original architecture, it was the most prominent in unsupervised techniques, scoring with accuracy (0.9975), precision (0.9890), recall (1.0000), F1-Score (0.9945), ROC-AUC (0.9989), FPR (0.0033) and FNR (0.0000). Its confusion matrix contains 180 true positives, 606 true negatives, 2 false positives and 0 false negatives.

Isolation Forest yielded accuracy (0.9530), precision (0.8488), recall (0.9667), F1-Score (0.9039) and ROC-AUC (0.9601). It found 174 of the 180 attacks but also reported 31 false positives. K-Means provided perfect precision but an FPR of zero. The recall at this point was still 0.2500, with only 45 of the 180 attacks being found. The F1-Score obtained was 0.4000 and indicates that K-Means is too risk averse.

This finding was also reflected by the ROC curves represented in Figure 4.2. The autoencoder had an ROC-AUC of 0.9989 and its ordering on attacks is extremely close to optimal. Isolation Forest offered a reasonable capability of discrimination too, and had

ROC-AUC of 0.9601. On the contrary, K-Means had an ROC-AUC of 0.8950 but does not allow enough recall for detecting an attack.

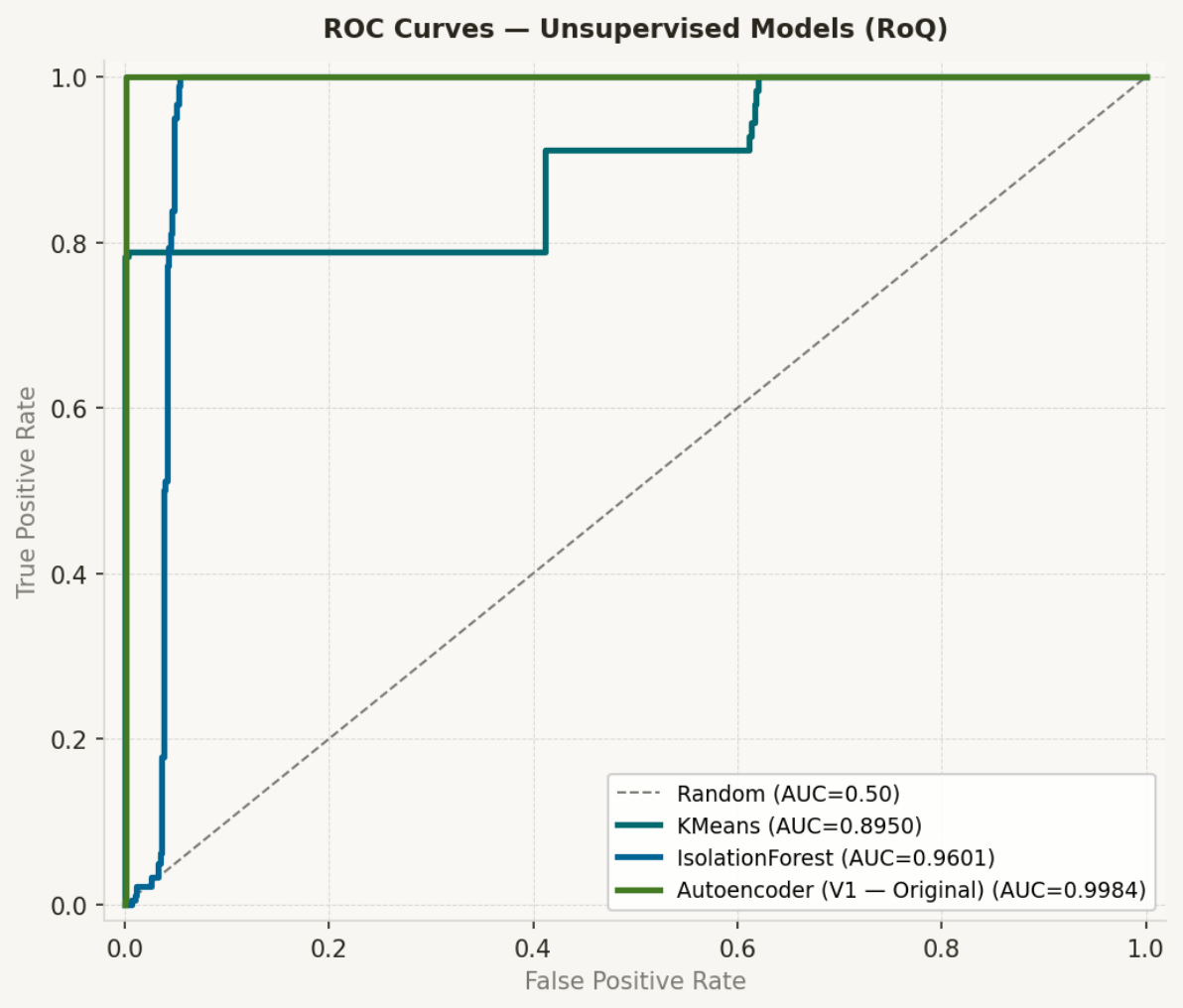


Figure 4.2: ROC curves for unsupervised ML detection on RoQ Laboratory Dataset.

The PCA also supported to the previous assumption. As could be seen in the PCA plot in Figure 4.3, the RoQ flows did not present themselves as one single tightly knit group that could be completely partitioned using K-Means. Their abnormal nature was better brought forth using reconstruction error and randomized partitioning. Consequently, autoencoders appear to be a better option for learning the normal behaviour manifold and Isolation Forest a possible replacement foranomaly detection.

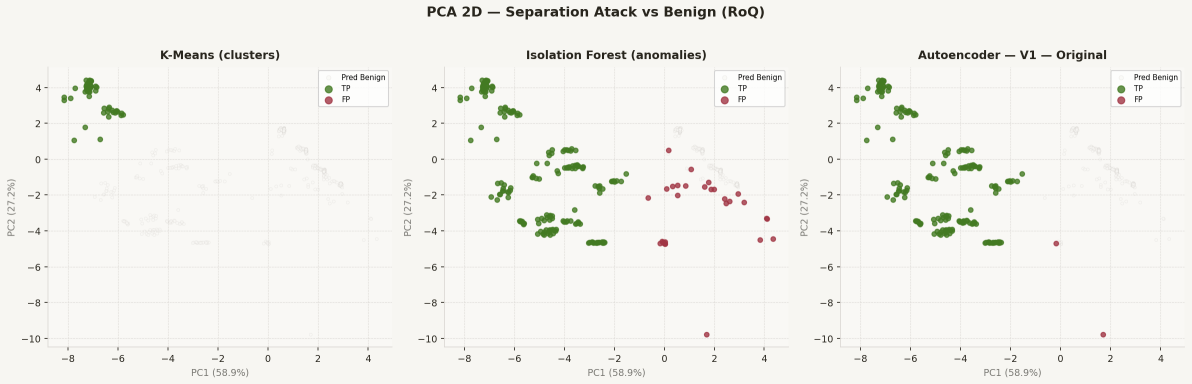


Figure 4.3: PCA plot for RoQ Laboratory Dataset.

4.5.4.1 K-Means Cluster Selection

Both the Elbow method and the Silhouette Score were examined to determine the ideal number of clusters, as shown in Figure 4.4. Within cluster sum of squares declined with the number of clusters but the Silhouette score had an optimal at $k = 3$, where the score was ≈ 0.646. Thus $k = 3$ was chosen.

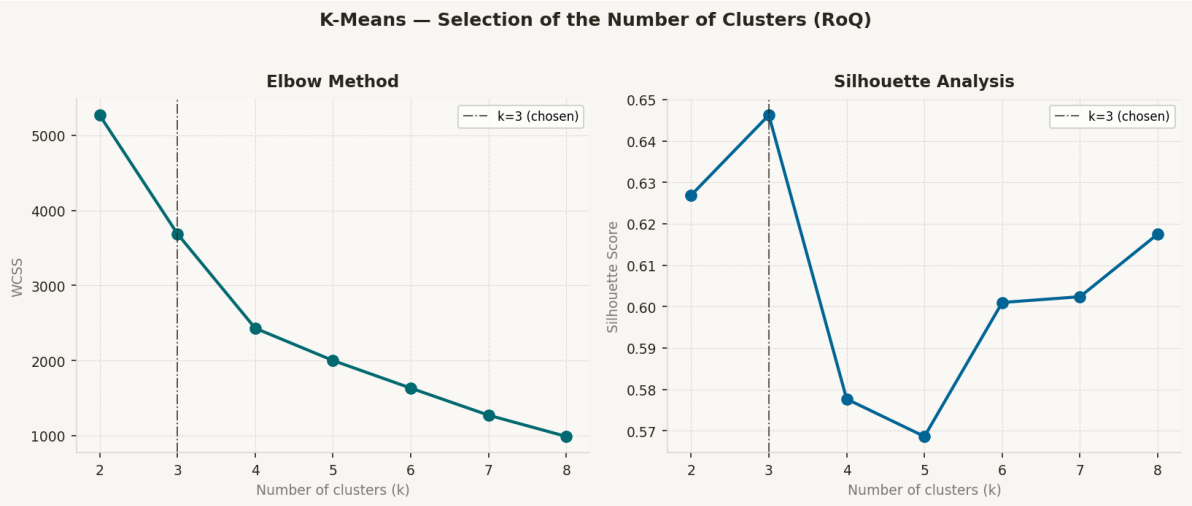


Figure 4.4: K-Means clustering results for RoQ detection (Elbow + Silhouette).

Despite the fact that this configuration has the unsupervised clustering criteria validation support, it failed to produce significant accuracy performance. K-Means generated no false positives, but fail to report 135 out of 180 samples associated to attacks. As it could be pointed out, cluster compactness is not the same with class detection performance. A cluster solution may be internally valid, but failed in grouping all attack pattern towards the anomalous class.

4.5.4.2 Isolation Forest Contamination Sweep

The fraction which was assumed to have an anomaly is shown by the contamination parameter. Table 4.5 shows example points of the sweeping mechanism performed by the Isolation Forest.

Contamination	Precision	Recall	F1
0.02	0.2500	0.0222	0.0408
0.10	0.7089	0.3111	0.4324
0.18	0.8239	0.6500	0.7267
0.22	0.8391	0.8111	0.8249
0.24	0.8466	0.8889	0.8672
0.26	0.8439	0.9611	0.8987
0.28	0.8145	1.0000	0.8978
0.40	0.5714	1.0000	0.7273
0.50	0.4569	1.0000	0.6272

Table 4.5: Representative results from Isolation Forest contamination sweep on RoQ Laboratory Dataset.

For low contaminant values the model is very conservative and has low recall. With contamination the recall increases, but the precision starts to lower, at contamination = 0.26 we get the best F1-Score, with F1-Score = 0.8987 (with precision of 0.8439 and recall of 0.9611).

Looking at this sweep shows that the contamination value is a functional rather than pure technical hyperparameter. If one desires to catch all attacks values of contamination 0.28 or more have recall (1.0000), although this is with reduced precision. The value 0.26 appears to be a reasonable trade off with good recall close to one at considerably higher values for precision.

4.5.4.3 Autoencoder Threshold Sensitivity and Architecture Comparison

When comparing reconstruction error distributions of benign traffic and attack traffic, the separation is clear, as the distribution of the reconstruction error values for benign flows falls very close to 0, whereas the reconstruction error for attack flows is much higher. The maximum-F1 threshold for V1 is 0.0995, and was identified a second threshold of 0.2558 so that the recall would remain ≥ 0.95.

Three kinds of autoencoding techniques were tried and compared:

- V1 - Original: Uses the full set of features;
- V2 - Top Features: Selects the relevant features;
- V3 - BN + Dropout: Uses batch normalization with dropout.

They are compared in Table 4.6 in two working points.

Variant	Threshold criterion	Accuracy	Precision	Recall	F1	AUC	FPR	FNR	TP	TN	FP	FN
V1 - Original	Max-F1	0.9975	0.9890	1.0000	0.9945	0.9989	0.0033	0.0000	180	606	2	0
V1 - Original	Recall ≥ 0.95	0.9975	0.9944	0.9944	0.9944	9989	0.0016	0.0056	179	607	1	1
V2 - Top Features	Max-F1	0.9860	0.9424	1.0000	0.9704	0.9971	0.0181	0.0000	180	597	11	0
V2 - Top Features	Recall ≥ 0.95	0.9860	0.9471	0.9944	0.9702	0.9971	0.0164	0.0056	179	598	10	1
V3 - BN + DP	Max-F1	0.9924	0.9780	0.9889	0.9834	0.9986	0.0066	0.0111	178	604	4	2
V3 - BN + DP	Recall ≥ 0.95	0.9924	0.9888	0.9778	0.9832	0.9986	0.0033	0.0222	176	606	2	4

Table 4.6: Comparison of Autoencoder variants under maximum F1-Score and recall-constrained thresholds.

V1 (Original) seems the best choice, at both points of operation, such that it performed perfectly on recall (no FNs) at max F1-Score, and gives only 2 FPs. If FPs need to be reduced and 1 FN is acceptable, it has 1 FP instead of 2. V2 (Top Features) had a considerably higher FPR and proves that feature selection is detrimental to obtaining an informative signature to reconstruct an anomaly. V3 (BN+DP) was not superior over V1, neither regarding F1-Score or errors.

This result was quite significant as it seems to indicates that the most informative parts of the RoQ information are spread across the entirety of engineered features. Though the feature importance calculation can select a subset of those most relevant features, the autoencoder benefits from incorporating the complementary pieces of information available in other features.

4.5.5 Summary of Observed Results

The overal summary between models is in Table 4.7. The supervised analysis gave KNN and SVM a 100% accuracy for this stratified 5-fold cross validation set. RF, MLP, and AdaBoost were very close at a F1-Score of 0.9973. For the unsupervised task, the autoencoder V1 performed at a F1-Score of 0.9945 without using any of the malicious traffic.

Category	Model	F1	AUC	FPR	FNR	Main Observations
Supervised	KNN	1.0000	1.0000	0.0000	0.0000	Perfect cross-validation result
Supervised	SVM	1.0000	1.0000	0.0000	0.0000	Perfect cross-validation result
Supervised	RF	0.9973	1.0000	0.0016	0.0000	Strong, interpretable baseline
Supervised	AdaBoost	0.9973	1.0000	0.0016	0.0000	Strong boosting alternative
Unsupervised	Autoencoder V1	0.9945	0.9989	0.0033	0.0000	Best unsupervised detector
Unsupervised	Isolation Forest	0.9039	0.9601	0.0510	0.0333	Strong but parameter-sensitive
Unsupervised	K-Means	0.4000	0.8950	0.0000	0.7500	Conservative, but misses most attacks

Table 4.7: Best-performing models and principal operating characteristics for RoQ detection.

The bottom line is that the RoQ traffic is very separable over the considered feature space, but.supervised and unsupervised should not be read in a similar way. The supervised one uses attack labels and thus acts as a target for our classification results. Since the autoencoder provides a comparable F1-Score while not using any labels for its training it's a more applicable solution if we want to imagine what would happen in case of a new kind of attack would appear.

4.6 Conclusions

Within this chapter, i developed, coded and evaluated a flow-level RoQ detection framework employing both supervised and unsupervised ML algorithms. The experiments were ran on the OVHCloud, where i used both individually generated benign and attack traffic streams that were then converted to flow-level features.

The supervised test, presented herein showed the chosen feature set classified RoQ and Benign traffic equally well within stratified 5-fold cross validation. The results for the supervised test were KNN and SVM perfect. It appeared that, for the other classifiers the F1-Score is 0.9973 for AdaBoost, RF, and MLP respectively. This suggested that the RF baseline is particularly compelling due to its combination of both optimal performance and interpretability.

On the unsupervised side, the best method was autoencoder V1-Original where F1-Score reached 0.9945, ROC-AUC reaches 0.9989 while FPR was 0.0033 and FNR was 0.0000. The Isolation Forest was a good complement, especially to K-means or when there was dependence on contamination parameter, and the K-means will not be effective enough as the sole detector by yielded 0 false positives while at the expense of 75% of attacks being not detected.

The feature importance demonstrates that the detection of RoQ mainly depended on `burst_cv`, `bytes_rate_cv`, `burst_peak_ratio`, `tcp_ratio`, `pkt_size_mean` and `pkt_size_std`. These were what i expected of the detection of RoQ since not just the packet count was required but the characteristics of the attack's impact must be presented via variations of burst and byte rate, spread of packet size and ratios at TCP level.

The key identified drawbacks were the controlled nature of the tested dataset, the susceptibility of anomaly detection to the selection of the threshold, the contamination, and the fact that further testing would be needed on non-homogeneous traffic distributions. However, the results confirmed that the reduced flow-based representation was effective in detection of RoQ traffic and that anomaly detection based on autoencoders can be considered a valid solution if attack information isn't known.

5

Hybrid Detection Model for RoQ Attacks

5.1 Introduction

Up till now supervised learning techniques have been compared against unsupervised learning techniques separately in the context of stealthy DoS attack detection, mainly focusing on slowloris and RoQ traffic. For RoQ, the experiment demonstrated supervised techniques are overall better discriminators than unsupervised ones while anomaly oriented approaches are still beneficial to complement existing mechanisms to distinguish from legitimate traffic.

Among all the supervised classifiers assessed within Chapter 4, KNN and SVM offered the best combination of accuracy and functional stability for detection purposes. As seen with the best F1-Score and, as important, the zero FPR and zero FNR, both of these classifiers excelled, and the classification was stable across all the folds. At the same time, the literature review identified that hybrid detection methods using a combination of ML, fuzzy logic, and distance-based reasoning are suitable for stealthy attacks because stealthy attacks create uncertain and marginal traffic that can only be discriminated by having multiple models and robust boundaries, rather than simply one definitive model or hard thresholds.

Motivated by this consideration, this chapter defines and assesses a Hybrid Detection Model specifically for RoQ traffic. The hybrid approach was thoroughly examined only for RoQ, not only due to RoQ being one of the core attack families under examination in this dissertation, but also since the finding in Chapter 4 demonstrated that RoQ is the most relevant scenario to evaluate whether or not a hybrid approach could offer a benefit in terms of operating-point-level tradeoff.

5.2 Motivation and Design Rationale

The rationale behind the hybrid approach arises directly from the RoQ results developed previously. Even though the top supervised classifiers worked quite well, the RoQ detection problem is more uncertain than the Slowloris one because the feature space demonstrates significant overlap between malicious and bursty benign traffic. This overlap means that

the score of one classifier may not necessarily provide enough information for a solid decision especially in the operating areas near the class boundary.

Thus, this new hybrid model was formulated by integrating these three signals, complementary in nature. A supervised or unsupervised classifier outputs a class score, the Euclidean distance based technique outputs how close each sample is from a strong centroid representative of benign and of attacks, and a fuzzy inference mechanism that aggregates these two signals to output the final score.

This design is highly applicable to the domain of Intrusion detection, where in practice, unnecessary alarms have operational costs, and thus decreasing false alarms is highly desirable. It may well be the case that a detector that only slightly under-performs on the global F1-Score, but significantly under-performs on the FPR is more useful in practice as it is easier to use for the analyst. Thus, the goal is not to find the supervised detector that out-performs on all measures, but rather, can the error profile be manipulated to one that is more desirable, and can this be justified statistically.

5.3 Proposed Hybrid Detection Model

5.3.1 Hybrid Architecture

The proposed hybrid detector is a score fusion-based architecture building upon the RoQ feature representation already developed in Chapter 4. The 14 window-level features used previously are also included here, comprising the packet count, the total number of bytes, the ratio of TCP packets, statistics over packet sizes, burstiness indicators and metrics on the packet inter-arrival times ensuring a direct comparison to previous supervised and unsupervised RoQ experiments.

The whole dataset was initially separated into training, validation and testing set using stratification. Normalization of the features was conducted using `StandardScaler`, fitted on training data set and then transformed on validation and test sets, which avoided the data leaking problem. Once the features have been normalized, strong centroids of the benign and attack class were calculated from the training data set using interquartile range filtration, so that the geometric references (used by distance component) are not corrupted by outliers.

After that, the normalized distance score was calculated as a relative indicator of how close each sample is to the benign and attack centroids, again for each sample. The score did not serve as a classification itself but was a piece of supporting geometric information that helped support whether a traffic window is near benign behavior or near attack-like behavior. For stealthy RoQ traffic, the additional piece of information helped in identifying structural deviation that might not be wholly captured by the classifier probability.

5.3.2 Candidate Base Models

The hybrid system did not make the central search on all models we tested, but only on a small, most relevant group of candidates. The choice was made to maintain methodological consistency with the general logic of benchmarking used in this dissertation, and to make the space of a hybrid search not too huge.

For this first comparison phase, four potential hybrid versions were tested:

- $Fuzzy + SVM + Distance$;
- $Fuzzy + KNN + Distance$;
- $Fuzzy + RF + Distance$;
- $Fuzzy + AE + Distance$.

SVM and KNN were taken because they yielded the highest supervised F1-Score in the RoQ study, Random Forest because it gave the best balance and consistency overall and the autoencoder because it was the most suitable unsupervised anomaly-oriented baseline.

This choice was due to the positioning the unsupervised models have throughout the dissertation. They were not treated as a real supervised detection alternative, but instead as complementing baselines that demonstrate whether using anomaly-oriented behaviour would bring additional information to a hybrid system.

5.3.3 Decision Fusion Mechanism

A fuzzy inference system processed two normalized inputs, such as the classifier score and the distance score in order to determine the fusion result. The classifier score was split into fuzzy partitions low, medium, and high confidence, and the distance score was split into the same partitions. The inputs were then linked to attack likelihood ranges from benign to very high attack by means of a set of 9 understandable fuzzy rules:

1. If classifier score was high and distance score was low → attack likelihood is very high;
2. If classifier score was high and distance score was medium → attack likelihood is high;
3. If classifier score was medium and distance score was high → attack likelihood is high;
4. If classifier score was medium and distance score was medium → attack likelihood is medium;
5. If classifier score was low and distance score was low → attack likelihood is low;

6. If classifier score was low and distance score was medium → attack likelihood is medium low;

7. If classifier score was medium and distance score was low → attack likelihood is medium low;

8. If classifier score was high and distance score was low → attack likelihood is medium;

9. If classifier score was low and distance score was high → attack likelihood is medium.

The fuzzy system combined rules like, (classifier score was High) and (distance evidence was High) then (attack was strong), and (classifier score was Low) and (distance evidence was Low) then (attack was benign). The middle ranges, such as when the classifier score was medium were mapped into soft output regions, thereby giving a much more natural representation of uncertainty than that offered by a hard-threshold detector. This resulting continuous hybrid score was then thresholded on the validation set, using an operating point that maximized F1-Score.

This architecture allowed a desirable level of interpretation, and importantly fited into the empirical benchmarking framework of the previous chapters. Specifically, it meant that the hybrid detector could be examined as an explicit variation of the RoQ classifiers of previous chapters, instead of a new, completely black-box system.

5.4 Experimental Methodology

5.4.1 Initial Hybrid Comparison

The first step of the evaluation was to compare between the main hybrid variants in a hold-out test scenario. The data was divided into a train, validation and test sets by stratified sampling and each hybrid variant was trained and optimized on the same feature set and scaling technique. Each fuzzy threshold was optimized using the validation set for each model and then tested in the test set. The tests performed to compare the models were `accuracy`, `precision`, `recall`, `F1-Score`, `ROC-AUC`, `FPR` and `FNR`.

This initial step was actually only for screening, in order to evaluate which base model was appropriate for more advanced hybrid assessment, not to be the ultimate validation protocol. Indeed, one train-validation-test split was more susceptible to the effects of the specific partition compared to multiple cross-validations.

5.4.2 Repeated Stratified K-Fold Evaluation

For the second stage of evaluation, the SVM hybrid branch was chosen for deeper examination since not only initial hybrid outcomes, but also stable and reliable results with balanced performances of SVM classifier derived from chapter 4 justified such selection. Therefore, there are four SVM based branches to carry out the second stage evaluation:

- $Fuzzy + SVM + Distance$;
- $Fuzzy + SVM$;
- $SVM + Distance$;
- SVM Only.

The different variants were tested with 5 times stratified 10-fold cross-validation resulting in 50 paired observations per metric and per variant. With this approach, could get a more reliable estimation on the variation of the results and, importantly, can test the variants statistically and withpairedtests.

During this phase, not only was the total prediction quality of interest, but also the functional trade-off between FPR and FNR. This interest is not unreasonable, given the function this hybrid detector was intended to perform, that was an attempt to see if the RoQ detector's error profile can be shifted towards less false alarms, without sacrificing attack detection abilities altogether.

5.4.3 Statistical Significance Analysis

Paired Wilcoxon signed-rank test was used on the fold level results to assess whether the discrepancies seen between the SVM baseline and hybrid versions were significant. The Wilcoxon test was suitable here as the data was paired across the same repeats of the folds and do not make any strong assumption of normality [7].

Besides p-values, Cliff's Delta also quantified the direction and practical size of effects reported, adding to the statisticial significance in that it indicates, in non-parametric terms, that the difference was negligible, small, medium, or large [25]. As several pairwise comparisons had been made correction procedures were also necessary. Holm-Šídák was applied for the main hypotheses of FPR and FNR [14] and Benjamini-Hochberg was used for the secondary hypotheses of F1-Score, ROC-AUC, precision and recall [3].

The distinction between the primary and secondary hypothesis was theoretically necessary. It was primarily motivated by the expected practical benefits derived from the hybrid model for the purpose of avoiding false alarms and missed detections. It should be stated that the other measures were not unimportant, however they were subservient to the main inquiry in this chapter.

5.5 Observed Results

5.5.1 Initial Comparison of Hybrid Variants

The initial comparison of the four hybrid variants is shown in Table 5.1.

Variant	Accuracy	Precision	Recall	F1	AUC	FPR	FNR	TP	TN	FP	FN	Threshold
$Fuzzy + SVM + Distance$	0.9747	1.0000	0.8889	0.9412	0.8898	0.0000	0.1111	32	122	0	4	0.36
$Fuzzy + KNN + Distance$	0.9684	0.9697	0.8889	0.9275	0.9440	0.0082	0.1111	32	121	1	4	0.05
$Fuzzy + RF + Distance$	0.9494	0.8889	0.8889	0.8889	0.9422	0.0328	0.1111	32	118	4	4	0.31
$Fuzzy + AE + Distance$	0.9367	1.0000	0.7222	0.8387	0.7234	0.0000	0.2778	26	122	0	10	0.36

Table 5.1: Initial comparison of the hybrid variants for RoQ detection.

Accuracy and F1-Score of 0.9747 and 0.9412, respectively were obtained with the highest values with the approach of $Fuzzy + SVM + Distance$ where precision is equal to 1.0000 and FPR equal to 0.0000. However it was found that the recall of this was 0.8889 with 4 false negatives.

The second highest F1-Score was achieved by $Fuzzy + KNN + Distance$ and its value was 0.9275 while the ROC-AUC value was the best and it was found to be equal to 0.9440 with 1 false positive and 4 false negatives. $Fuzzy+RF+Distance$ on the other hand yielded 0.8889 and 0.9422 with 4 false positives and 4 false negatives. $Fuzzy + AE + Distance$ achieved best values for both precision and FPR with 1.0000 and 0.0000 respectively, but its recall was low and was found to be 0.7222, indicating 10 false negatives.

This initial examination revealed that for a single test $Fuzzy + SVM + Distance$ best balances precision and recall, $Fuzzy + KNN + Distance$ showed the greatest classification ability by ranking using ROC-AUC, however this generated one false positive, $Fuzzy + RF + Distance$ performed less accurately than the SVM and KNN versions of this system, while $Fuzzy + AE + Distance$ proved too cautious, failing to identify any but the smallest group of these attacks.

The confusion matrices corroborated these findings. $Fuzzy+SVM+Distance$ correctly classified the 122 benign traffic streams and 32 of 36 attack streams, $Fuzzy + KNN + Distance$ incorrectly classified only one benign traffic streams while it correctly classified 32 of 36 attack traffic streams, $Fuzzy + RF + Distance$ recorded 4 false positives and $Fuzzy + AE + Distance$ recorded 0 false positives but missed 10 attacks for the latter.

This confirmed a general trend identified earlier in Chapter 4, where the best performing RoQ hybrid models seem to be those which incorporate the best possible supervised classifier, and those of a solely anomaly based detection type seem to be more of supporting points of reference than as core detection tools.

Though, the hold-out results provided slightly better stability of the behaviour with the SVM based hybrid the deviation between the SVM and KNN branch was not large enough to draw any conclusive decision on the single split alone. Therefore, RF branch was selected to be tested on repeated cross-validation since it represents a good combination between highest values of the hybrid screening with the overall most consistent behaviour.

5.5.2 RSKF Evaluation of SVM-Based Variants

Figure 5.1 summarizes the results of the repeated stratified K-fold evaluation for the SVM-based variants. The heatmap shows the mean values of F1-Score, ROC-AUC, FPR and FNR across the 50 paired observations for each variant.

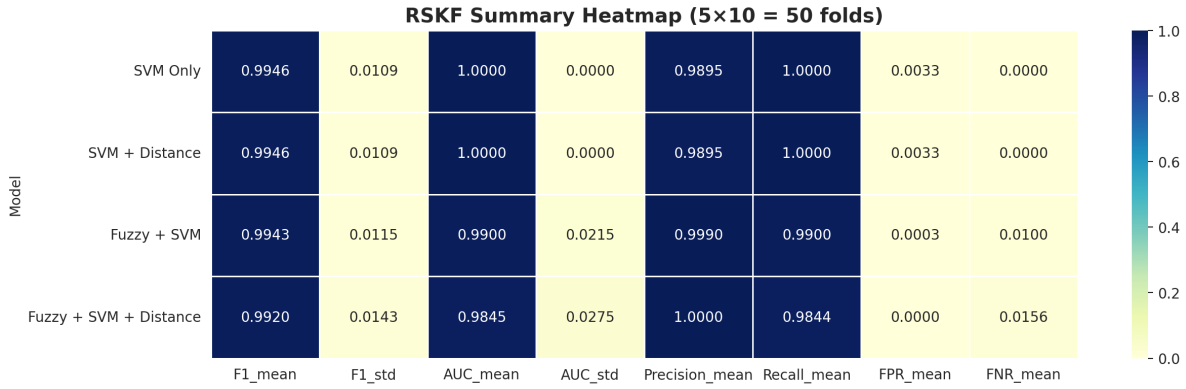


Figure 5.1: Summary of the results of the repeated stratified K-fold evaluation for the SVM-based variants.

By utilizing the repeatedly stratified K-fold cross validation, the behavior of the hybrid model was made clear.

The average metric for SVM Only and $SVM + Distance$ were exactly the same, F1 mean was 0.9946, ROC-AUC mean was 1.0000, precision mean was 0.9895, recall mean was 1.0000, FPR mean was 0.0033 and FNR mean was 0.0000.

$Fuzzy + SVM$ decreased mean FPR to 0.0003 and increased mean precision to 0.9990, but mean recall decreased to 0.9900 and FNR mean increased to 0.0100.

$Fuzzy + SVM + Distance$ didn't missed any negative example, means FPR was 0.0000 and mean recall decreased further, from 0.9900 to 0.9844, and mean FNR increased from 0.0100 to 0.0156. Its F1 mean and ROC-AUC mean were 0.9920 and 0.9845, respectively.

Consistently across these outcomes, the findings demonstrated that there was indeed a functional trade off. The addition of the fuzzy elements had decreased the false positives but also increase the false negatives. The strongest increase in FPR was for the distance feature when added to a $Fuzzy + SVM$, but it also achieved the worst recall and the largest dip in F1-Score and ROC-AUC.

5.5.3 Wilcoxon and Effect Size Analysis

The Wilcoxon signed-rank test was applied to the fold-level results to assess the statistical significance of the differences observed between the SVM Only baseline and the other variants. The results of this analysis are summarized in Table 5.2.

Metric	Variant	Baseline mean	Variant mean	Mean difference	p_{corr}	Cliff's δ	Significant
FPR	$Fuzzy + SVM$	0.00331	0.00033	-0.00298	0.00973	+0.182	✓
FPR	$Fuzzy + SVM + Distance$	0.00331	0.00000	-0.00331	0.00585	+0.200	✓
FPR	$SVM + Distance$	0.00331	0.00331	0.00000	1.00000	0.000	×
FNR	$Fuzzy + SVM$	0.00000	0.01000	+0.01000	1.00000	-0.180	×
FNR	$Fuzzy + SVM + Distance$	0.00000	0.01556	+0.01556	1.00000	-0.260	×
F1	$Fuzzy + SVM$	0.99459	0.99432	-0.00028	1.00000	-0.036	×
F1	$Fuzzy + SVM + Distance$	0.99459	0.99197	-0.00263	1.00000	-0.112	×
AUC	$Fuzzy + SVM$	1.00000	0.99002	-0.00998	1.00000	-0.180	×
AUC	$Fuzzy + SVM + Distance$	1.00000	0.98448	-0.01552	1.00000	-0.260	×
Precision	$Fuzzy + SVM$	0.98947	0.99895	+0.00947	0.01172	+0.180	✓
Precision	$Fuzzy + SVM + Distance$	0.98947	1.00000	+0.01053	0.01172	+0.200	✓
Recall	$Fuzzy + SVM$	1.00000	0.99000	-0.01000	1.00000	-0.180	×
Recall	$Fuzzy + SVM + Distance$	1.00000	0.98444	-0.01556	1.00000	-0.260	×

Table 5.2: Results of the Wilcoxon signed-rank test comparing the SVM Only baseline with the other variants.

Regarding to the first evaluation metric, the FPR values both obtained statistical significant decreases after the application of Holm-Šidák correction method. Both $Fuzzy + SVM$ (from 0.00331 to 0.00033, $p_{corr} = 0.00973$) and $Fuzzy + SVM + Distance$ (from 0.00331 to 0.00000, $p_{corr} = 0.00585$) presented very small effects, with the following Cliff's δ: 0.182, 0.200.

Similarly to accuracy, $Fuzzy + SVM$ and $Fuzzy + SVM + Distance$ shown significant improvement by using Benjamini-Hochberg correction. The precision was increased from 0.98947 up to 0.99895 in $Fuzzy + SVM$ and up to 1.00000 in $Fuzzy + SVM + Distance$ with $p_{corr} = 0.01172$.

The reduction in recall and the increase in FNR were statistically insignificant post correction. Likewise F1-Score and ROC-AUC were not significant post Benjamini-Hochberg correction. Based on corrected findings a specific operational benefit was proposed the use of fuzzy inference, with or without distance, does decrease false positive and increase precision but does not improve the resulting F1-Score or recall.

The uncorrected p-values were more optimistic and indicated the same trend. FPR were significantly reduced while precision was significantly increased for $Fuzzy + SVM$ and $Fuzzy + SVM + Distance$ before correction. Correction was used for final evaluation as multiple variants were experimented and multiple metrics were reported.

From the paired fold differences it was possible to observe that FPR decrease was in line for both fuzzy variants, but FNR increase was for a fewer number of folds. That is why the FNR comparisions did not get the priority but the FNR main comparisions stold get significant post correction.

5.5.4 Summary of Observed Results

The key findings of the analysis are summarized in Table 5.3.

Evaluation	Model	F1	AUC	Precision	Recall	FPR	FNR	Main Conclusions
Initial comparison	$Fuzzy + SVM + Distance$	0.9412	0.8898	1.0000	0.8889	0.0000	0.1111	Best F1 and zero FPR
Inital comparison	$Fuzzy + KNN + Distance$	0.9275	0.9440	0.9697	0.8889	0.0082	0.1111	Best ROC-AUC
Initial comparison	$Fuzzy + RF + Distance$	0.8889	0.9422	0.8889	0.8889	0.0328	0.1111	More false positives
Initial comparison	$Fuzzy + AE + Distance$	0.8387	0.7234	1.0000	0.7222	0.0000	0.2778	Precise but misses attacks
Repeated evaluation	SVM Only	0.9946	1.0000	0.9895	1.0000	0.0033	0.0000	Strong baseline
Repeated evaluation	$Fuzzy + SVM$	0.9943	0.9900	0.9990	0.9900	0.0003	0.0100	Significant FPR reduction
Repeated evaluation	$Fuzzy + SVM + Distance$	0.9920	0.9845	1.0000	0.9844	0.0000	0.0156	Maximum FPR reduction

Table 5.3: Summary of the key findings from the initial comparison and repeated stratified K-fold evaluation of the SVM-based hybrid variants.

For the first comparison the best one was $Fuzzy + SVM + Distance$ on F1-Score and FPR, and $Fuzzy + KNN + Distance$ on ROC-AUC. With the second comparison of SVM fuzzy variants, FPR and precision were enhanced dramatically on, however recall and FNR also dropped sharply.

The above findings could thus be interpreted not as general improvement across all metrics but as a compromise in operation among approaches. The approach to be favored when the best approach to minimize FPR is sought was $Fuzzy + SVM + Distance$, $Fuzzy + SVM$ for less drastic reduction of FPR yet with greater recall value than that boosted approach, and SVM Only will serve better when total coverage against attack and highest F1-Score/ROC-AUC are desired.

5.6 Conclusions

In this chapter, the Hybrid Detection Model for RoQ attacks using supervised classification or unsupervised classification, fuzzy logic and distance based reasoning was proposed and demonstrated. It should be noted the re-adjusted initial comparison used as base models for SVM, KNN, RF and autoencoder.

For the initial evaluation report, can see $Fuzzy + SVM + Distance$ reaches the highest F1-Score (0.9412) and had no false positives. $Fuzzy + KNN + Distance$ had the highest ROC-AUC (0.9440) and $Fuzzy + AE + Distance$ had 0 false positives but had a recall of only 0.7222. It is clear from this that the base classification method plays a significant role in the operating trade off of the hybrid model.

The repeated SVM based analysis was the one which had the principal statistics in my results. $Fuzzy + SVM$ brings mean of FPR downed form 0.00331 to 0.00033 and precision up from 0.98947 to 0.99895. $Fuzzy + SVM + Distance$ bring FPR downed to 0.00000 and precision up to 1.00000. Both those change were significant under all multiple comparison correction.

On contrary, both variation lowered down and raised up, respectively recall and FNR (not statistically significant after correction), and, at a lesser extent, slightly decreased F1-Score and ROC-AUC, again not significantly different.

So the amended results justified an operating benefit of the hybrid solution, where fuzzy inference contributed an operating benefit, decreasing false positive alarms and therefore increasing the precision but having an impact that doesn't helped overall detectection

coverage. The distance element will further increase the effect of FPR reduction, yet be responsible for the greatest reduction in recall. A compromise must be reached according to the objective of the application, for maximum recall, achieve by SVM Only for its optimal F1-Score/ROC-AUC value, for balanced false alarm reduction achieve by the combined $Fuzzy + SVM$, and for minimum FPR achieve by $Fuzzy + SVM + Distance$.

The main limitation here is that the chosen thresholds and achieved trade offs were evaluated using a single traffic distribution. In future studies, hybrid solutions should be implemented across more OVHCloud instances, and varying degrees of traffic, background traffic and RoQ intensity, in order to ascertain whether the reduced FPR holds in other contexts.

6

Conclusions and Future Work

6.1 Main Conclusions

This dissertation focused on the detection of stealthy and low-rate DoS attacks, specifically Slowloris and RoQ. An experimental environment has been designed and implemented on OVHCloud instances that allowed to generate and capture benign and malicious traffic in a controlled environment and to extract flow temporal, statistical, burst and TCP features. These features have been tested with supervised learning, anomalous behavior detection based on unsupervised algorithms and combined fuzzy and distance-based detection methods.

Overall, the results indicated that the flow-level features used were sufficient to detect both attacks in all scenarios. The models for Slowloris performed well on the controlled dataset and were also very highly ranked for the external IFTO dataset, with the autoencoder providing the best unsupervised alternative. For RoQ, the supervised models performed well across the board while the autoencoder provided nearly perfect unsupervised detection. For the hybrid evaluation, it was found that fuzzy SVM variants dramatically lowered the FPR and raised the precision over the SVM only baseline, but came at the expense of a lower recall and higher FNR. Therefore, the hybrid model did not uniformly improve on all metrics, but instead provided a useful operational trade-off when prioritizing a reduction in the number of false alerts.

Overall, the research showed that stealthy DoS attacks were best detected in a detection pipeline that reflects their temporal and behavioral characteristics.

6.2 Future Work

Though encouraging results were achieved, the assessment was perfromed in a laboratory environment with a limited number of traffic scenario and models, indicating room for future work:

6.2.1 Extension of the Experimental Evaluation

The dimensions of future experiment evaluation studies should include a larger range of workloads, longer capture durations, greater variation in traffic generation and a wider variety of benign workloads. Controlled traffic generation conditions were used in these experiments, which was appropriate for reproducibility, although these do not represent all the conditions in a production network.

Further, it is suggested that the testbed evolve from a single attacker instance to the distributed attack paradigm of multiple coordinated attackers. This scenario is assumed to be conducive to an evaluation of the ability of the proposed features and models to generalise to attack traffic coming from different sources, places and rates. However, more experiments are suggested with different Slowloris connection numbers, header intervals, RoQ burst periods, attack strengths, legitimate traffic levels and background noise. Examining the models in different OVHCloud regions, instance types and network constructs might better support conclusions on their ability to generalise.

6.2.2 Evaluation of Additional Models

As seen in the literature, detecting Slowloris and RoQ attack by means of unsupervised machine learning is uniquely challenging. With that in mind, it should be considered to evaluate the following approaches for the unsupervised classification, such as Multi-Autoencoder Architecture, Isolation Forest with Dynamic Time Warping, One-Class Support Vector Machine with RBF Kernels and Unsupervised Packet-Level Neural Models.

6.2.3 Use of Additional Statistical Techniques

Statistical techniques are expected to offer important future work directions for detection of Slowloris and RoQ attacks. With that in mind, it should be considered to evaluate the use of additional statistical techniques, such as Information Theoretic Metrics and Entropy, Time-Series Analysis and Statistical Smoothing, Advanced Distance Metrics and Multidimensional Outliers and Self-Similarity and Fractal Analysis.

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